

**INTEGRATIVE ANALYSIS OF PD-L1 ASSOCIATED
IMMUNE GENE EXPRESSION AND TUMOUR
MICROENVIRONMENT IN TRIPLE NEGATIVE BREAST
CANCER**

*Progress Report submitted for the partial fulfillment of the degree of
Bachelor of Technology in Biotechnology*

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TABLE OF CONTENTS:

Abstract.....	8
1. Introduction and Review of Literature.....	9
2. Literature Review.....	10
3. Aim and objective.....	14
5. Materials and Methods.....	15
6. Results and Discussion.....	18
7. Conclusion.....	32
8. Future Directions.....	33
9. References.....	33
Annexure A: Full DEG Tables.....	35
Annexure B: CIBERSORTx Immune Cell Boxplots.....	38
Annexure C: R Analysis Pipeline.....	50
Annexure D: Supplementary Figures.....	67
DrillBit Plagiarism Check Report.....	68

LIST OF FIGURES:

Figure No.	Caption.....	Page
Fig 1	Definition of Triple-Negative Breast Cancer (Credits: NCI).....	10
Fig 2	The Tumor Immune Microenvironment in TNBC (Yang et al.).....	11
Fig 3	Therapeutic Approaches to Target Specific Pathways in TNBC (Wu et al.).....	12
Fig 4	Distribution of Breast Cancer Intrinsic Subtypes within the TCGA-BRCA Cohort.....	18
Fig 5	PCA of Basal TNBC (Showing PD-L1 High and PD-L1 Low).....	19
Fig 6	PCA Heatmap of Basal TNBC.....	20
Fig 7	Volcano Plot Showing Differentially Expressed Genes across TNBC Samples.....	21

Fig 8	Volcano Plot of DEGs between PD-L1–High and PD-L1–Low TNBC Samples.....	22
Fig 9	Distribution of Tumor Mutational Burden (TMB) in the TCGA-BRCA Cohort.....	23
Fig 10	Spearman Correlation between PD-L1 Expression and Immune Cell Fractions in TNBC	24
Fig 11	Bubble Plot Showing Significance and Strength of PD-L1 vs Immune Infiltration.....	25
Fig 12	Kaplan-Meier Survival Analysis by CD8 Tcell infiltration.....	26
Fig 13	KM Survival Analysis stratified by TRS using binary grouping.....	27
Fig 14	KM Survival Analysis stratified by TRS using tertile grouping.....	27
Fig 15	Spearman Correlation Heatmap of Key Immune and Molecular Variables.....	28
Fig 16	KM survival analysis by TDO2 Expression.....	29
Fig 17	Scatter Plots of IDO2 Expression vs M2 Macrophages.....	30
Fig 18	Scatter Plots of IDO2 Expression vs Tregs.....	30
Fig 19	Upregulated DEGs.....	35-36
Fig 20	Downregulated DEGs.....	37
Fig 21	CD8 Tcells by Tumor Subtype	38
Fig 22	B Cells naive by Tumor Subtype.....	39
Fig 23	B Cells memory by Tumor Subtype.....	39
Fig 24	Plasma Cells by tumor Subtype.....	40
Fig 25	T cells CD8 by Tumor subtype.....	40
Fig 26	T Cells CD4 Naive by Tumor Subtype.....	41
Fig 27	T Cells CD4 Memory by Tumor Subtype.....	41
Fig 28	T Cells CD4 Memory Activated by Tumor Subtype.....	42
Fig 29	T Cells Follicular Helper by Tumor Subtyp.....	42
Fig 30	T Cells Regulatory (Tregs) by Tumor Subtype.....	43
Fig 31	T Cells Gamma Delta by Tumor Subtype.....	43
Fig 32	NK Cells Resting by Tumor Subtype.....	44
Fig 33	NK Cells Activated by Tumor Subtype.....	44
Fig 34	Monocytes by Tumor Subtype.....	45
Fig 35	Macrophages M0 by Tumor Subtype.....	45
Fig 36	Macrophages M1 by Tumor Subtype.....	46
Fig 37	Macrophages M2 by Tumor Subtype.....	46
Fig 38	Dendritic Cells Resting by Tumor Subtype.....	47
Fig 39	Dendritic Cells Activated by Tumor Subtype.....	47

Fig 40	Mast Cells Resting by Tumor Subtype.....	48
Fig 41	Mast Cells Activated by Tumor Subtype.....;	48
Fig 42	Eosinophils by Tumor Subtype.....	49
Fig 43	Neutrophils by Tumor Subtype.....	49
Fig 44	Voom Mean-Variance Trend Plot for TNBC Expression Data Prior to DEG Analysis.....	67
Fig 45	Stacked Barplot of Immune Cell Composition per Sample (CIBERSORTx).....	67

ABSTRACT

Triple-Negative Breast Cancer (TNBC) is an aggressive subtype characterized by the absence of ER, PR, and HER2, resulting in limited targeted therapeutic options and modest responsiveness to immune checkpoint inhibitors. TNBC remains challenging to treat due to the highly variable nature of its immune microenvironment. Although immune checkpoint inhibitors (ICIs) targeting the PD-1/PD-L1 axis have expanded therapeutic possibilities, only a subset of TNBC patients derive meaningful benefit. Conventional biomarkers such as PD-L1 levels, tumor mutational burden (TMB), and TIL scoring provide an incomplete picture when used in isolation due to biological heterogeneity, sampling variability, and inconsistent analytical standards. To address these limitations, this study employs an integrative transcriptomic framework using publicly available TCGA-BRCA data accessed via the UCSC Xena repository. A cohort of 76 Basal-like TNBC samples was analyzed by integrating RNA-sequencing data, immune cell deconvolution estimates, somatic mutation burden, and clinical metadata. Samples were stratified by PD-L1 (CD274) expression, and differential gene expression analysis was performed using the limma-voom framework. Immune cell composition was estimated using CIBERSORTx with the LM22 signature matrix, and Spearman correlations between PD-L1 and 22 immune cell populations were computed with Benjamini–Hochberg FDR correction. A novel composite Tumor-Reactive Score (TRS) was constructed by integrating z-scored PD-L1 expression, CD8⁺ T-cell fraction, and log-transformed TMB to capture the multidimensional immune state of each tumor. Additionally, IDO2 expression was extracted and correlated with immunosuppressive cell populations and Kaplan–Meier survival analysis was performed for multiple immune stratifications. The analysis identified a predominantly immune-activating transcriptomic signature in PD-L1-high tumors, with upregulation of genes including KLRC4, IFNG, IL21, and IDO2. CIBERSORTx deconvolution confirmed enrichment of activated CD8⁺ T cells and M1 macrophages in PD-L1-high tumors. IDO2 showed no significant correlation with Tregs or M2 macrophages, suggesting pathway-independent immune regulation. TRS-based survival stratification did not reach statistical significance in this cohort, likely reflecting sample size limitations. These findings collectively support the need for composite, multi-

parameter biomarker frameworks over single-marker approaches for guiding immunotherapy decisions in TNBC.

Keywords: TNBC, PD-L1, Spearman correlation, Kaplan Meier Survival analysis, TRS

1. INTRODUCTION AND REVIEW OF LITERATURE:

Triple-Negative Breast Cancer (TNBC) accounts for approximately 15–20% of all breast cancer diagnoses and is associated with poorer clinical outcomes due to the absence of hormone receptors (ER, PR) and HER2 amplification. This molecular profile limits targeted therapeutic options, positioning chemotherapy and, more recently, immune checkpoint inhibitors (ICIs) as primary systemic treatments. Although ICIs targeting the PD-1/PD-L1 pathway have demonstrated transformative efficacy in several malignancies, their benefit in TNBC remains modest, with objective response rates ranging from 8–20%.

The inconsistent therapeutic efficacy reflects the biological complexity of the TNBC tumor immune microenvironment (TIME), which displays striking variability in immune cell composition, spatial organization, and functional state. While features such as high tumor-infiltrating lymphocytes (TILs), CD8⁺ T-cell activity, and immune-related gene signatures indicate potential for immunogenic responses, parallel immunosuppressive influences—including tumor-associated macrophages, regulatory T cells, and alternative checkpoint pathways—frequently undermine therapeutic success.

Existing biomarkers used clinically (PD-L1 expression, tumor mutational burden, TIL density) are insufficient when evaluated independently, largely due to analytical variability, spatial heterogeneity, and limited predictive accuracy. These challenges highlight the need to shift from single-parameter biomarkers toward integrative models capable of capturing the multidimensional immune-biology of TNBC.

This review provides a comprehensive synthesis of the TNBC immune landscape, evaluates current and emerging biomarkers, and discusses the growing rationale for composite predictive frameworks to guide precision immunotherapy.

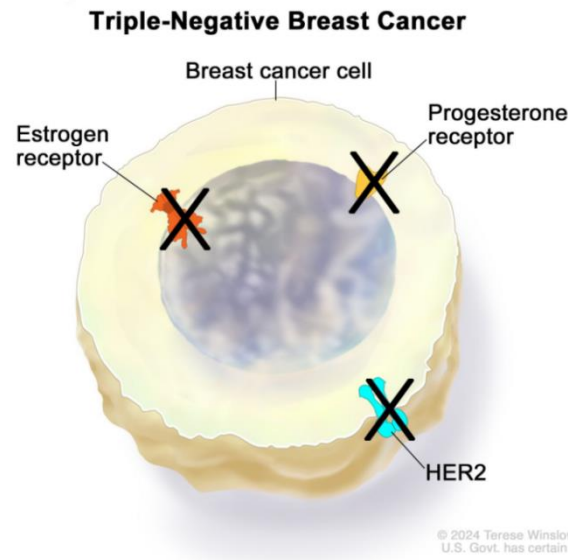


Fig 1 : Definition of triple- Negative Breast Cancer (Credits: NCI)

2. LITERATURE REVIEW

2.1. The Tumor Immune Microenvironment in TNBC

TNBC is distinguished by a more immunogenic TIME relative to other breast cancer subtypes. Many tumors display high infiltration of tumor-infiltrating lymphocytes (TILs), particularly CD8⁺ cytotoxic T cells and B cells, which correlate with favorable outcomes [1]. However, this immune-rich phenotype is not uniform across patients or even within individual tumors. Spatial analyses reveal both “immune-hot” regions, characterized by dense TIL infiltration, and “immune-cold” areas with sparse immune presence [2];[4]. Such heterogeneity contributes to inconsistent immunotherapy responses.

In addition, the TIME is shaped by immunosuppressive cells, including tumor-associated macrophages (TAMs), regulatory T cells (Tregs), and myeloid-derived suppressor cells, which collectively inhibit antitumor immunity [7]. Cytokines such as TGF- β and IL-10 further reinforce immune suppression, creating barriers to PD-1/PD-L1 blockade efficacy. These findings underscore the dual nature of the TIME in TNBC—as both a driver of immune activation and a source of resistance.

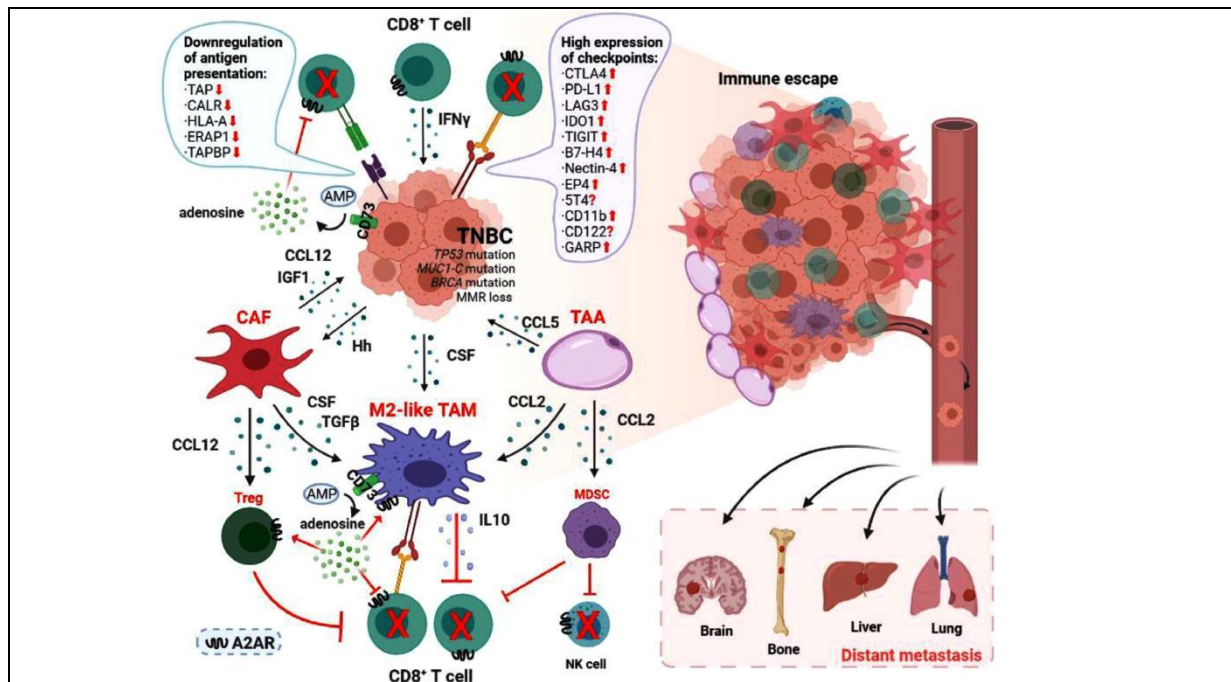


Fig 2 : The tumor immune microenvironment in TNBC (Yang et. al)

2.2. Biomarkers for Immunotherapy Response

2.2.1 PD-L1 Expression

PD-L1 expression, the most widely studied biomarker for ICIs, is currently used clinically to stratify TNBC patients. However, its predictive value is limited by methodological inconsistencies. Different assays (e.g., 22C3 vs SP142) yield variable positivity rates, and intratumoral heterogeneity further complicates interpretation [6]. Importantly, some PD-L1–negative patients still respond to ICIs, while PD-L1–positive cases may not, highlighting its inadequacy as a standalone marker.

2.2.2 Tumor Mutational Burden (TMB)

TMB has been proposed as a surrogate for neoantigen load and immunogenicity. In TNBC, higher TMB correlates with increased immune infiltration and response potential, but studies show inconsistent predictive power [5]. Moreover, TNBC typically exhibits intermediate TMB compared to melanoma or lung cancer, limiting its utility as a robust biomarker.

2.2.3 Tumor-Infiltrating Lymphocytes (TILs)

TIL density is a strong prognostic marker in TNBC, with higher levels linked to improved survival and therapeutic response. However, variability in scoring methods, reliance on manual pathology, and failure to capture functional diversity of immune cells constrain its application [1]. Emerging computational and deep learning approaches to TIL quantification aim to standardize assessment [2].

2.2.4 Immune Gene Signatures

Beyond single markers, immune gene signatures offer integrative perspectives. The immunologic constant of rejection (ICR), ferroptosis-related signatures, and macrophage enrichment profiles have been investigated for their predictive value in TNBC [3];[2]. These signatures capture complex immune processes that may underlie resistance or sensitivity to checkpoint blockade.

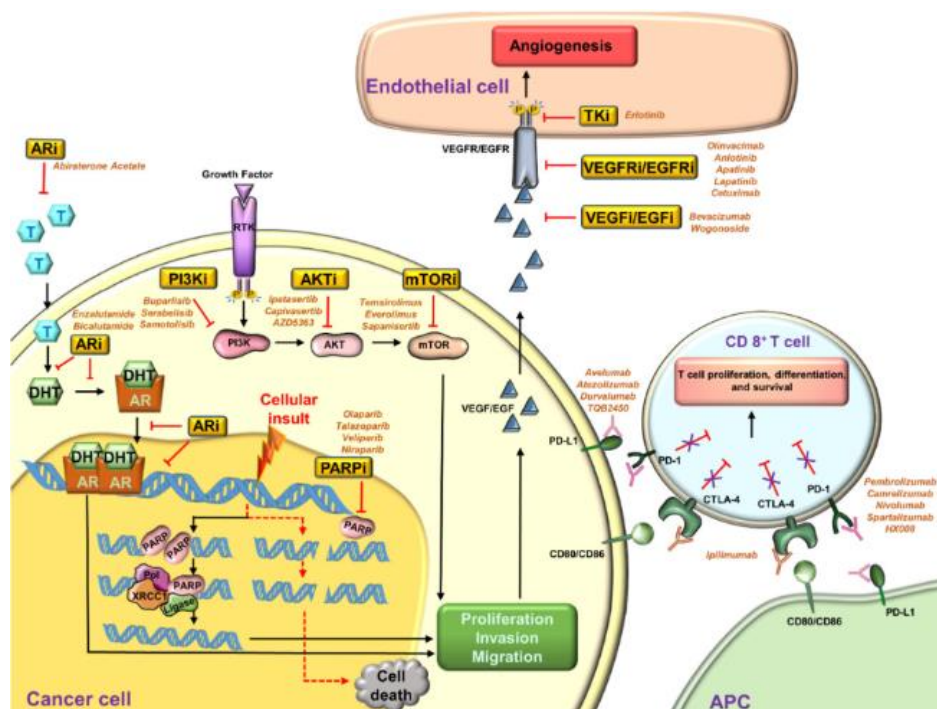


Fig 3: Therapeutic approaches to target specific pathways in TNBC (Wu et. Al)

2.3. Challenges and Knowledge Gaps

2.3.1 Heterogeneity of the TIME

The coexistence of immune-hot and immune-cold subregions within TNBC tumors complicates biomarker development. Even when PD-L1 or TILs are elevated in one region, other areas may remain immunologically silent, reducing overall treatment efficacy [4].

2.3.2 Immunosuppressive Pathways Beyond PD-1/PD-L1

While PD-1/PD-L1 blockade is central, resistance frequently emerges from alternative checkpoints (e.g., CTLA-4, VISTA) or immunosuppressive regulators such as UBR2 [7]. Neglecting these pathways risks oversimplifying the biology of ICI resistance in TNBC.

2.3.3 Limitations of Single Biomarkers

No single biomarker sufficiently predicts patient response to ICIs. PD-L1, TMB, and TILs each provide partial information, but their variability and lack of reproducibility limit clinical translation [6]; [5].

2.4. Toward Composite Biomarker Models

Growing evidence supports the integration of multiple biomarkers into composite models. Studies show that combining PD-L1 expression, TMB, TIL density, and immune signatures improves predictive accuracy compared to individual markers [8]. For example, ferroptosis-related gene signatures integrated with TIL scores stratify TNBC patients more effectively than PD-L1 alone [3]. Similarly, macrophage enrichment and adaptive immune resistance profiles provide additional granularity in predicting response to checkpoint blockade [2].

Clinical trials also reinforce the need for composite frameworks. The NeoSACT phase II study demonstrated that PD-L1 combined with immune infiltration scores more accurately predicted outcomes than PD-L1 alone in TNBC patients treated with neoadjuvant ICIs [5]. These findings emphasize that composite biomarkers could guide patient selection and optimize treatment strategies in clinical practice.

3. AIM AND OBJECTIVE:

To perform an integrative analysis of PD-L1-associated gene expression, immune microenvironment composition, and mutational landscape in Triple-Negative Breast Cancer (TNBC) using publicly available TCGA data, and to explore composite biomarker frameworks for refining immunotherapy response prediction.

- To collect and pre-process TNBC gene expression, immune, mutation, and clinical data from the UCSC Xena repository.
- To stratify TNBC samples by PD-L1 (CD274) expression and identify differentially expressed genes using the limma-voom framework.
- To estimate immune cell composition across TNBC samples using CIBERSORTx deconvolution with the LM22 signature matrix.
- To analyze Spearman correlations between PD-L1 expression and 22 immune cell populations with FDR correction.
- To construct a composite Tumor-Reactive Score (TRS) by integrating PD-L1 expression, CD8⁺ T-cell fraction, and log-transformed TMB as a multi-dimensional immune activation metric.
- To investigate the expression and immunosuppressive correlates of IDO2 in relation to Treg and M2 macrophage populations in TNBC.
- To evaluate the prognostic significance of immune infiltration, PD-L1 status, IDO2 expression, and TRS using Kaplan–Meier survival analysis.
- To visualize immune patterns using volcano plots, boxplots, heatmaps, bubble plots, and correlation matrices.

4. MATERIALS AND METHODS:

4.1. MATERIALS:

The following publicly available resources and tools were used:

4.1.1.Datasets

TCGA-BRCA (TNBC subset)

Gene expression (FPKM/TPM), clinical data, and mutation data obtained from UCSC Xena Browser.

4.1.2.Reference Gene Sets

LM22 immune signature matrix from CIBERSORTx.

4.1.3.Software & Platforms

R programming language (version 4.4.2)

Packages: UCSCXenaTools, ggrepel, maftools, openxlsx, limma-voom, tidyverse, dplyr, survival, survminer, pheatmap, ggplot2, reshape2.

4.1.4.CIBERSORTx web platform (for immune cell deconvolution).

4.1.5.Microsoft Excel / Google Sheets (for intermediate data handling)

4.2. METHODS:

4.2.1. Data Collection:

Triple-negative breast cancer (TNBC) samples were extracted from the TCGA-BRCA cohort using clinical subtype annotations from the UCSC Xena Browser. TCGA-BRCA clinical phenotype data was downloaded directly via the UCSCXenaTools R package querying the TCGA hub, using the dataset `TCGA.BRCA.sampleMap/BRCA_clinicalMatrix`. Gene expression data (log2-normalized RSEM counts) was retrieved from the HiSeqV2 dataset. Somatic mutation data was obtained from the BRCA MC3 MAF file hosted on the UCSC Xena S3 server. Gene expression (TPM/log2 TPM), clinical survival data, and sample metadata were downloaded.

4.2.2. Data Preprocessing:

TNBC samples were identified by triple-negative receptor status (ER-, PR-, HER2-) using the nature2012 annotations, yielding 123 samples, of which 76 were confirmed Basal-like by PAM50 classification and used for downstream analysis. Sample IDs were standardized to 12-character TCGA barcodes for cross-dataset merging.

Expression matrices were filtered for TNBC samples and normalized. Samples with incomplete clinical information were removed. PD-L1 (CD274) expression values were extracted for each sample.

4.2.3. PD-L1– based Stratification and PCA:

Samples were stratified into **PD-L1–high** and **PD-L1–low** groups using the median PD-L1 expression as the cutoff. To further assess whether PD-L1 expression influences global gene-expression structure, a Principal Component Analysis (PCA) was performed. PCA revealed a clear separation trend between PD-L1–high and PD-L1–low tumors along the first two principal components, indicating that PD-L1 expression is associated with broader transcriptomic shifts in the tumor microenvironment.

4.2.4. Differential Gene Expression Analysis:

Differential expression between PD-L1–high and PD-L1–low groups was performed using DESeq2/edgeR with thresholds of $|\log_2FC| > 1$ and $FDR < 0.05$.

4.2.5. TMB Distribution:

Tumor Mutational Burden (TMB) for breast cancer samples was obtained using the TCGA-BRCA somatic mutation (MAF) data. The MAF file was processed using the **maftools** R package, which computes TMB as the total number of somatic, non-synonymous mutations normalized per megabase (mut/Mb) of the exome. Only high-confidence variants (missense, nonsense, frameshift insertions/deletions, and splice-site mutations) were included. Samples with incomplete mutation profiles were excluded.

The final TMB values were summarized and visualized using a histogram to show overall distribution across the cohort.

4.2.5. Immune Cell Infiltration Analysis:

Normalized gene expression matrices were submitted to **CIBERSORTx** using the LM22 signature matrix to estimate the relative abundance of 22 immune cell types.

4.2.6. Integrated Correlation Analysis:

To capture the multidimensional immune landscape of TNBC, a composite Tumor Response Score (TRS) was constructed for each sample. PD-L1 expression (CD274), CD8⁺ T-cell fraction (from CIBERSORTx), and log2-transformed TMB were each independently z-score normalized and summed to generate the TRS. Samples were then stratified into Immune-High and Immune-Low groups using the median TRS as a cutoff. Spearman correlation analysis was performed between PD-L1 expression and all 22 immune cell populations estimated by CIBERSORTx, with p-values adjusted using the Benjamini–Hochberg FDR method. Correlation matrices and heatmaps were generated using the `pheatmap` package in R.

4.2.7. Survival Analysis:

Kaplan–Meier survival curves and Cox proportional hazards models were used to examine the association of PD-L1 and immune cell features with overall survival.

4.2.8. Visualization:

Plots including boxplots, heatmaps, barplots, and correlation matrices were generated in R (`ggplot2`, `pheatmap`) to summarize findings.

5. RESULTS:

5.1. Clinical Characteristics of the Cohort

The TNBC subset was successfully extracted from the TCGA-BRCA cohort using clinical subtype annotations. The resulting expression matrix showed substantial variability across tumor samples, providing a strong basis for downstream analyses. This dataset served as the foundation for dimensionality reduction, differential expression, and immune profiling.

Var1	Freq	Percent
	35	28.46
Basal-like	76	61.79
HER2-enrich	5	4.07
Luminal A	4	3.25
Luminal B	1	0.81
Normal-like	2	1.63

Fig 4: Distribution of breast cancer intrinsic subtypes within the TCGA-BRCA cohort.

5.2. PCA of TNBC samples

5.2.1 PCA Score Distribution

Principal Component Analysis revealed distinct variation patterns within the TNBC cohort. The PCA score plot demonstrated natural clustering tendencies among samples, indicating underlying transcriptional heterogeneity.

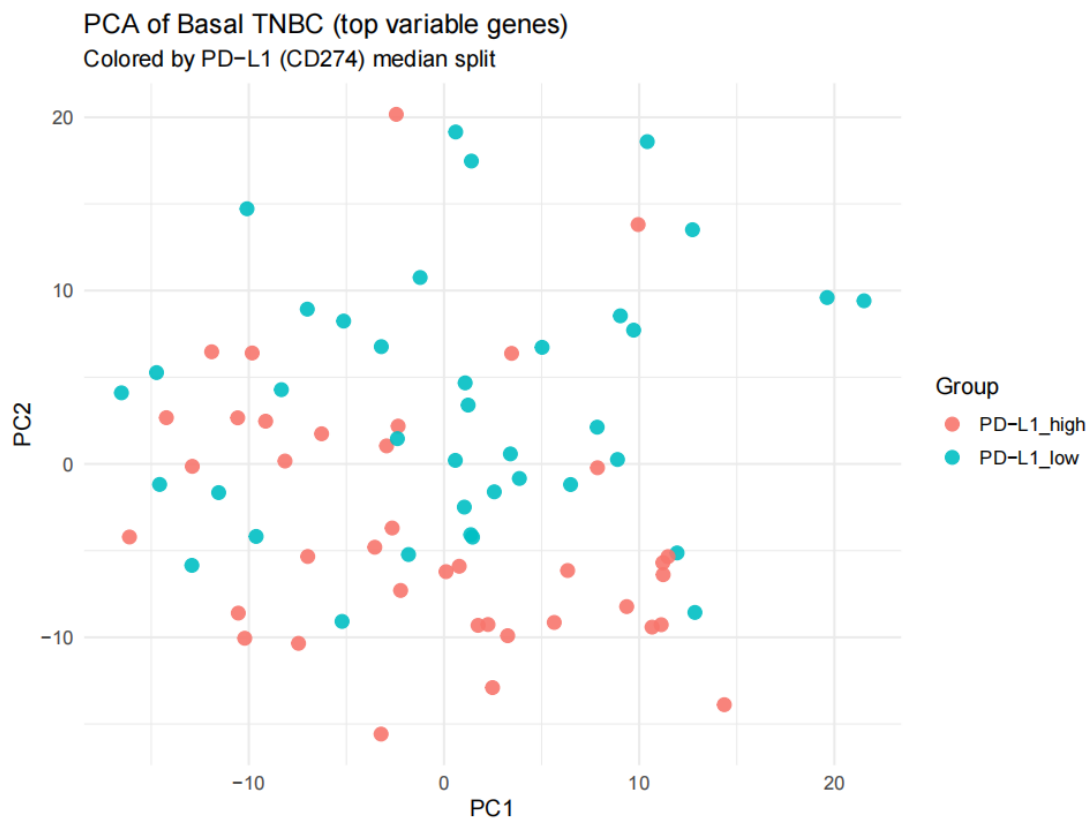


Fig 5: PCA of basal TNBC (Showing PD-L1 high and PD-L1 low)

5.2.2 PCA Heatmap

A heatmap generated using the top variable genes highlighted clear expression gradients across TNBC samples. These patterns further supported the variability observed in the PCA scores and justified the progression to differential gene expression analysis.

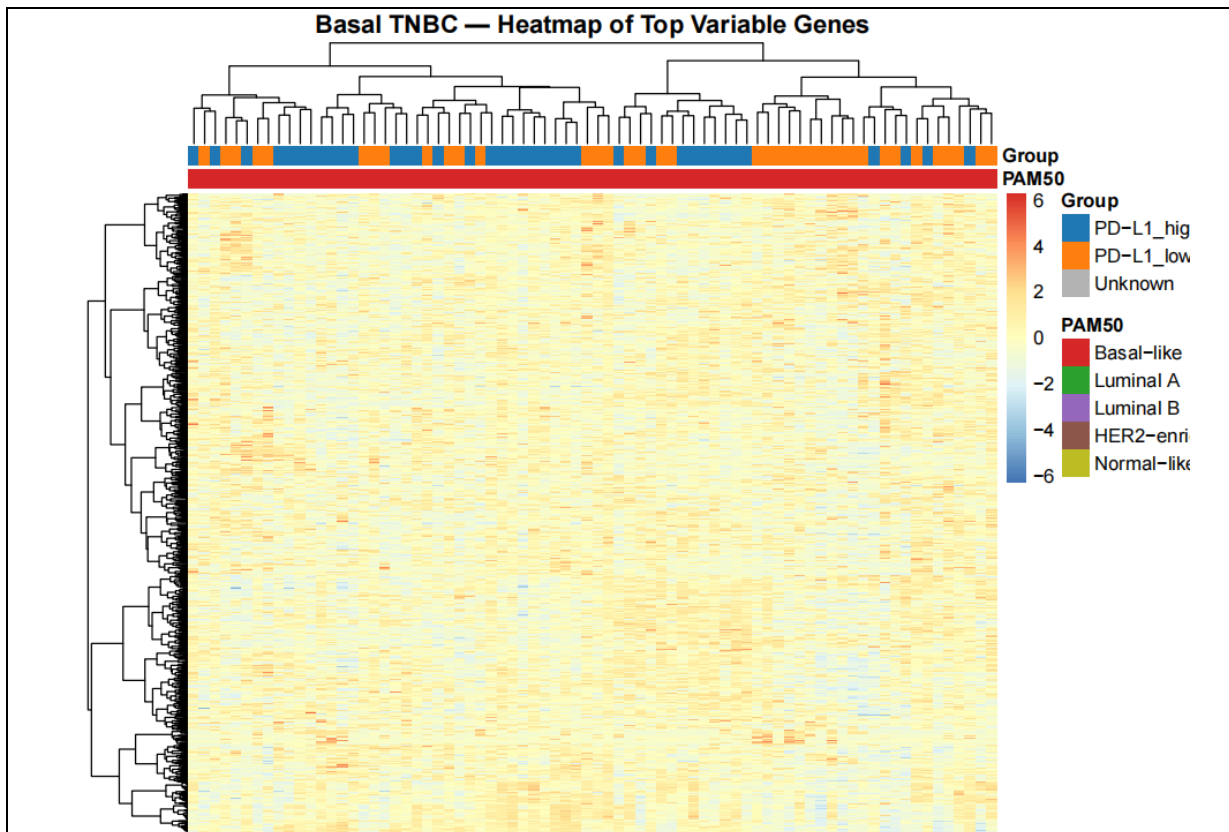


Fig 6: PCA Heatmap of Basal TNBC

5.3. Differential Gene Expression Analysis

5.3.1 Identification of Upregulated and Downregulated Genes

Differential expression testing between PD-L1–high and PD-L1–low groups produced well-defined sets of upregulated and downregulated genes. Many upregulated genes in PD-L1–high tumors were immune-associated, suggesting heightened immune activation, while downregulated genes included pathways indicative of suppressed metabolic or proliferative programs.

5.3.2 Volcano Plot of DEGs

A volcano plot summarizing global DEGs displayed a balanced distribution of significantly upregulated and downregulated genes, confirming robust differential expression. Significant transcripts showed strong fold changes and FDR-adjusted

significance.

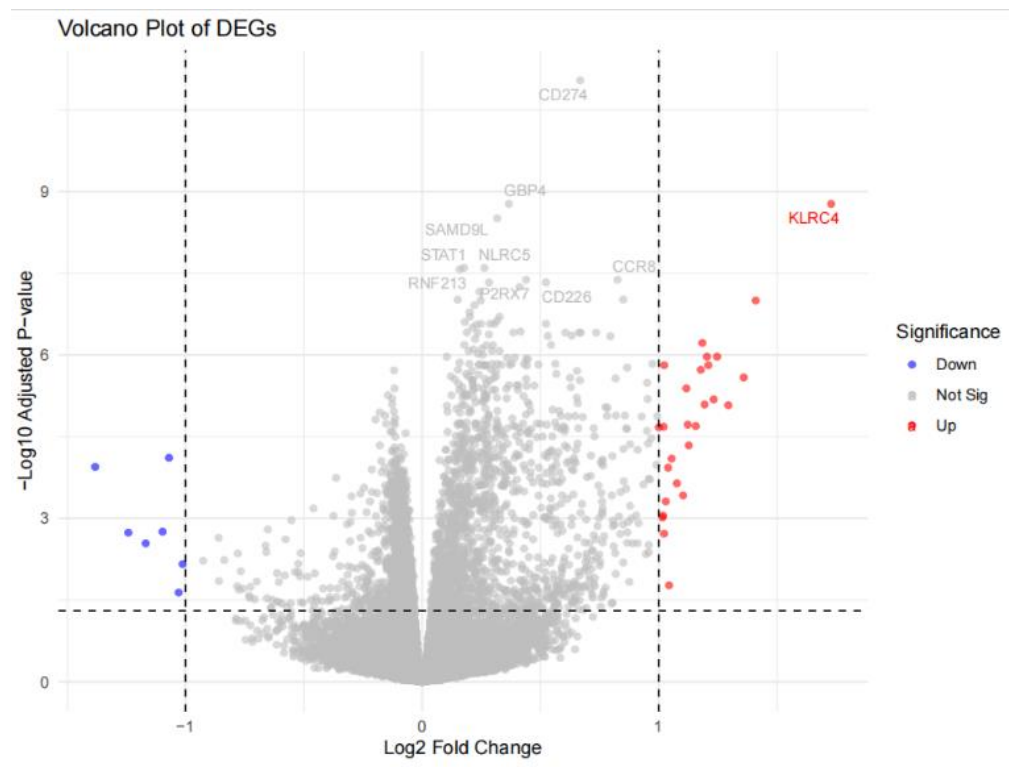


Fig 7: Volcano plot showing differentially expressed genes across TNBC samples.

Red points represent significantly upregulated genes, blue represent significantly downregulated genes ($|\log_2FC| > 1$, $FDR < 0.05$).

5.3.3. PDL1 specific DEG analysis

A second volcano plot focusing solely on PD-L1-linked DEGs identified genes most strongly associated with PD-L1 modulation. These genes included key immune regulators and stress-response mediators, underscoring PD-L1's influence on tumor-immune interactions.

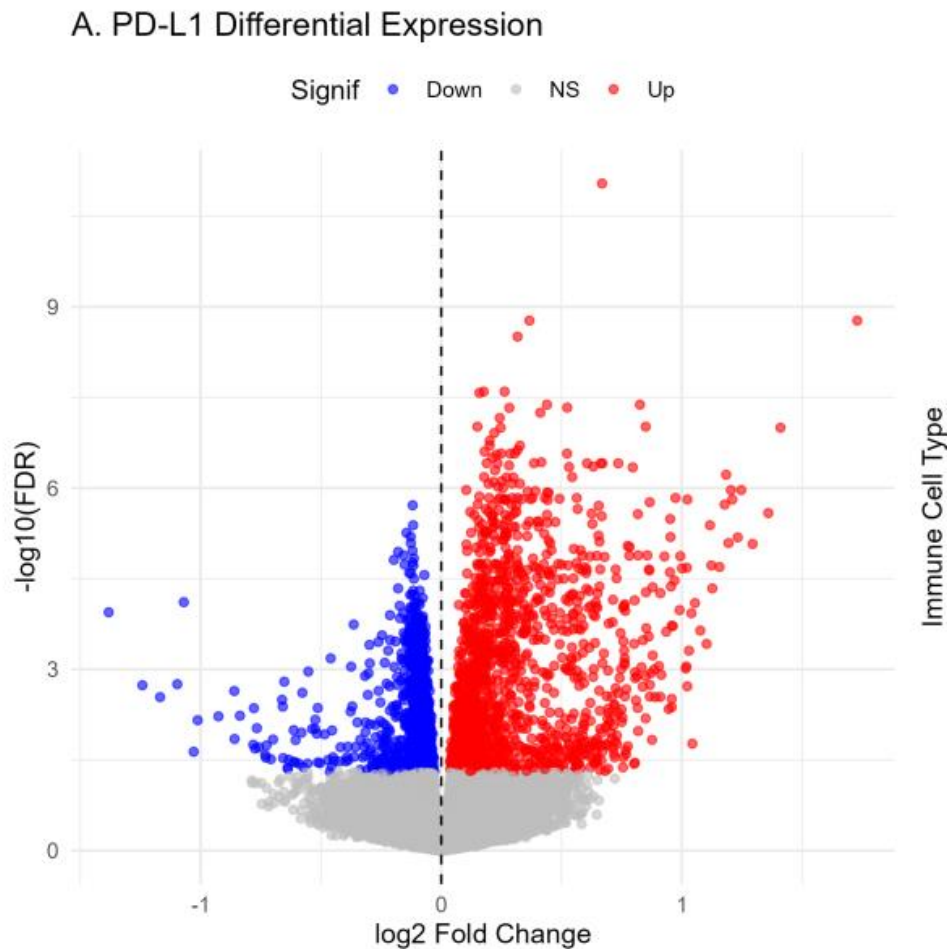


Fig 8: Volcano plot of differentially expressed genes between PD-L1–high and PD-L1–low TNBC samples. Upregulated genes in PD-L1-high tumors (red) include numerous immune-activation signatures, whereas downregulated genes (blue) correspond to pathways reduced in immune-cold tumors.

5.4. Tumor Mutational Burden Distribution in TNBC

Analysis of TMB across the TCGA-BRCA cohort showed that breast cancer tumors generally exhibit a low mutational burden, with the majority of samples clustered between 0–5 mutations/Mb. A small number of samples demonstrated higher TMB values, extending beyond 50 mutations/Mb, representing outliers with hypermutated phenotypes.

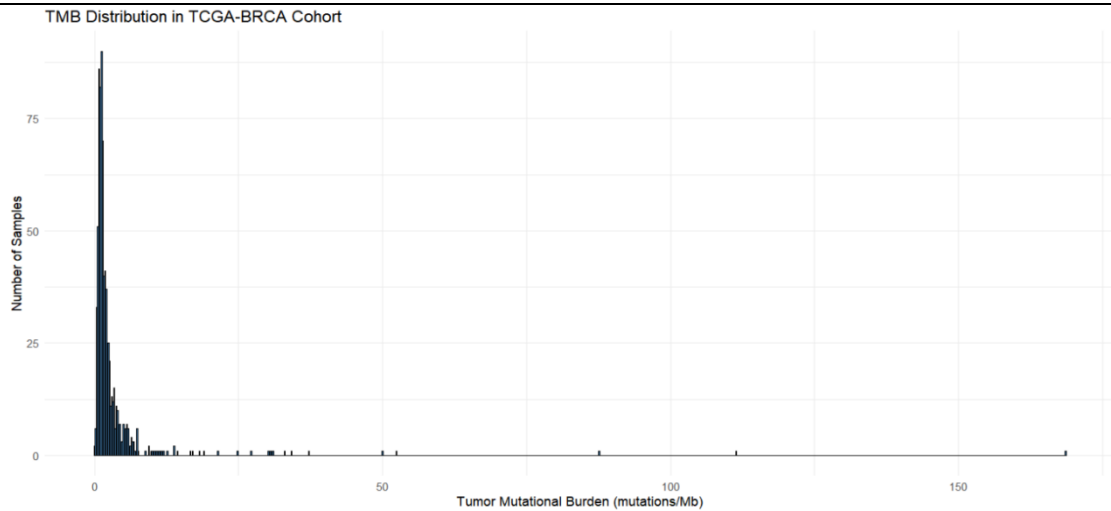


Fig 9: Distribution of Tumor Mutational Burden (TMB) in the TCGA-BRCA cohort.

5.5 Immune Cell Infiltration and CIBERSORTx Analysis

After establishing transcriptomic differences, immune deconvolution using CIBERSORTx provided a detailed map of immune infiltration patterns.

5.5.1 PD-L1 and Immune Cell Fractions

PD-L1 expression showed strong positive correlations with activated CD8⁺ T cells, M1 macrophages, and dendritic cell subsets. Conversely, negative correlations were observed with certain resting or suppressive immune populations.

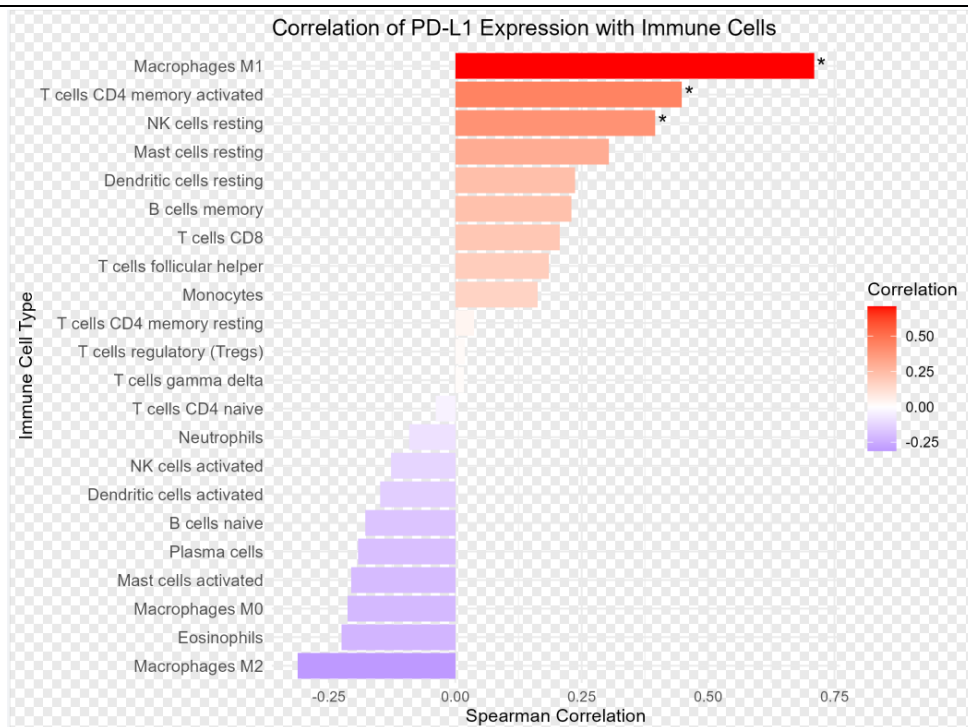


Fig 10: Spearman correlation between PD-L1 expression and immune cell fractions in TNBC. Positive correlations (red) indicate immune populations enriched in PD-L1–high tumors, while negative correlations (purple) represent cell types reduced in PD-L1–high tumors.

5.5.2 Immune Cell Expression Differences (Boxplots)

Boxplots comparing immune cell fractions between PD-L1–high and PD-L1–low samples revealed significant differences across multiple lineages, including CD8⁺ T cells, NK-cell subsets, macrophages, and T-helper cell categories.

5.5.3 Immune Correlation Networks

Correlation matrices showed highly coordinated immune behavior, including clusters of positively correlated effector cells and negatively correlated suppressive populations, reflecting a structured immune microenvironment in TNBC.

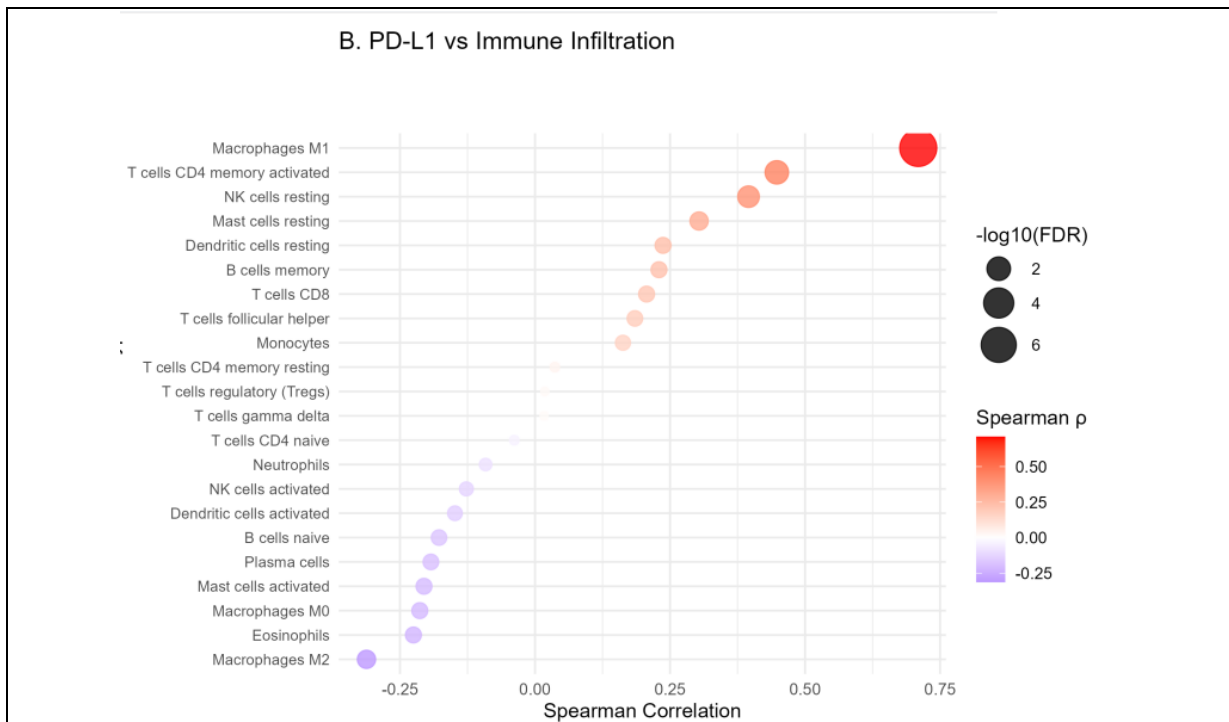


Fig 11: Bubble plot showing the significance and strength of association between PD-L1 expression and immune infiltration.

Bubble size reflects $-\log_{10}(\text{FDR})$, while color intensity represents the magnitude of the Spearman correlation.

5.6. Kaplan–Meier Survival Analysis (CD8⁺ T Cells)

Survival modeling demonstrated that higher CD8⁺ T-cell abundance corresponded with improved overall survival trends in TNBC patients. This supports the prognostic significance of cytotoxic T-cell infiltration and aligns with the immune-activated phenotype of PD-L1–high tumors.

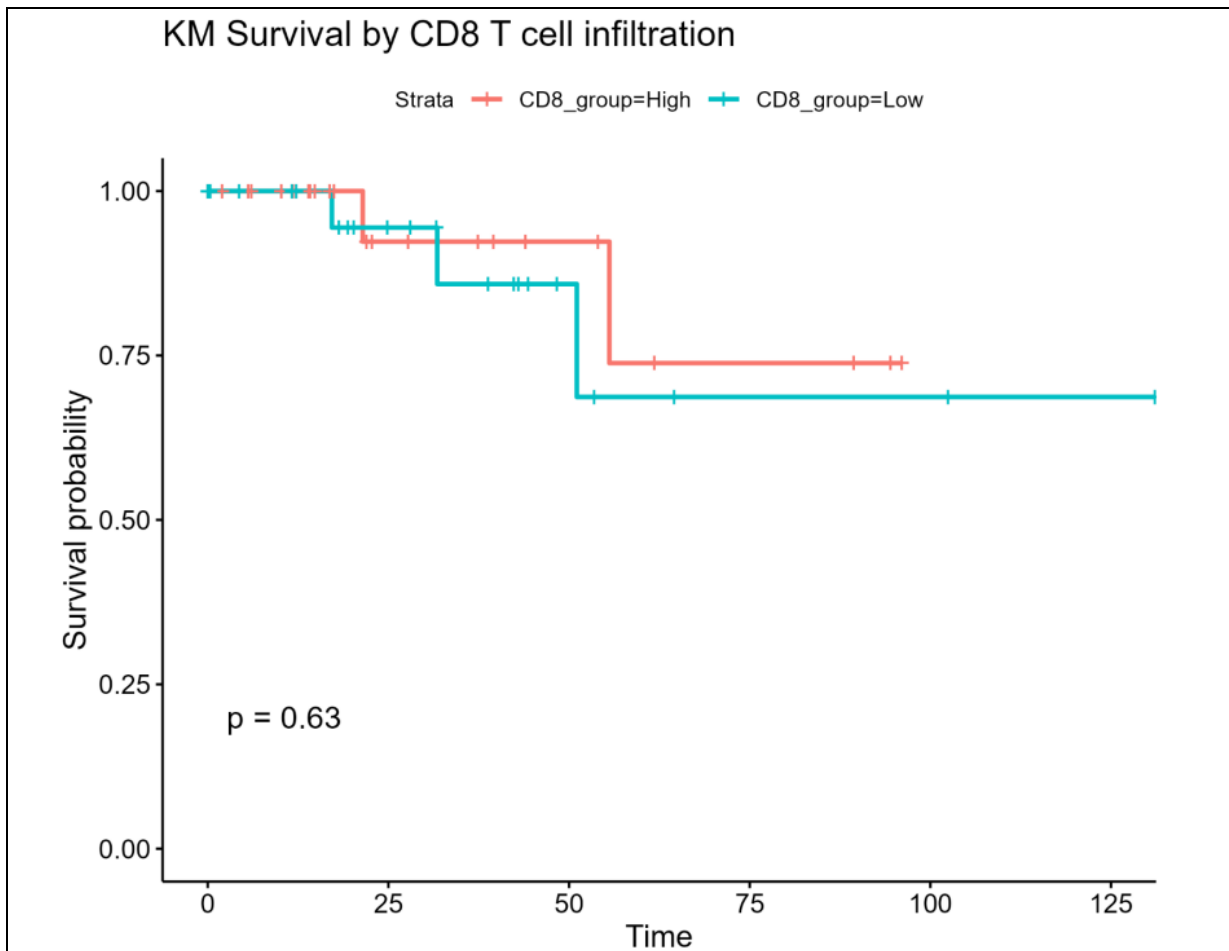


Fig 12: Kaplan-Meier Survival Analysis by CD8 Tcell infiltration

5.7. Data Integration and Sample Harmonization

CIBERSORTx immune cell fraction estimates, PD-L1 expression values (CD274 extracted from the expression matrix), TMB data, and TCGA clinical metadata were loaded and merged into a unified analytical dataframe. Sample IDs across all four datasets were standardized to the first 12 characters of the TCGA barcode format to ensure consistent matching. The final integrated dataframe included CD274 expression, TMB, CD8⁺ T cell fractions, overall survival time and event status, age at diagnosis, and pathologic stage.

5.8. Composite Tumor-Reactive Score (TRS)

A composite Tumor-Reactive Score (TRS) was derived by summing the z-scored values of PD-L1 expression, CD8⁺ T cell fraction, and log-transformed TMB. This approach integrates

three orthogonal dimensions of anti-tumor immunity into a single continuous score. Samples were then classified as Immune-High or Immune-Low based on the median TRS. Additionally, tertile-based grouping (Low, Medium and High) was applied for Kaplan-Meier visualization of TRS-stratified survival.

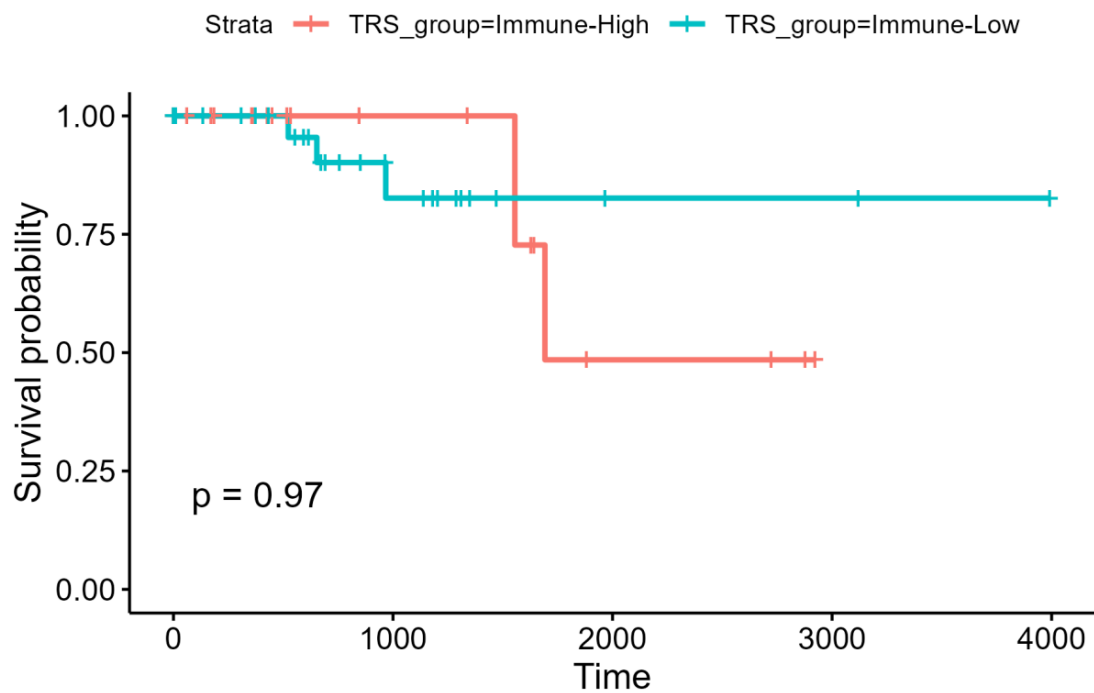


Fig 13: Kaplan–Meier Survival Analysis Stratified by Tumor Response Score (TRS) by Binary Grouping (Immune-High vs. Immune-Low)

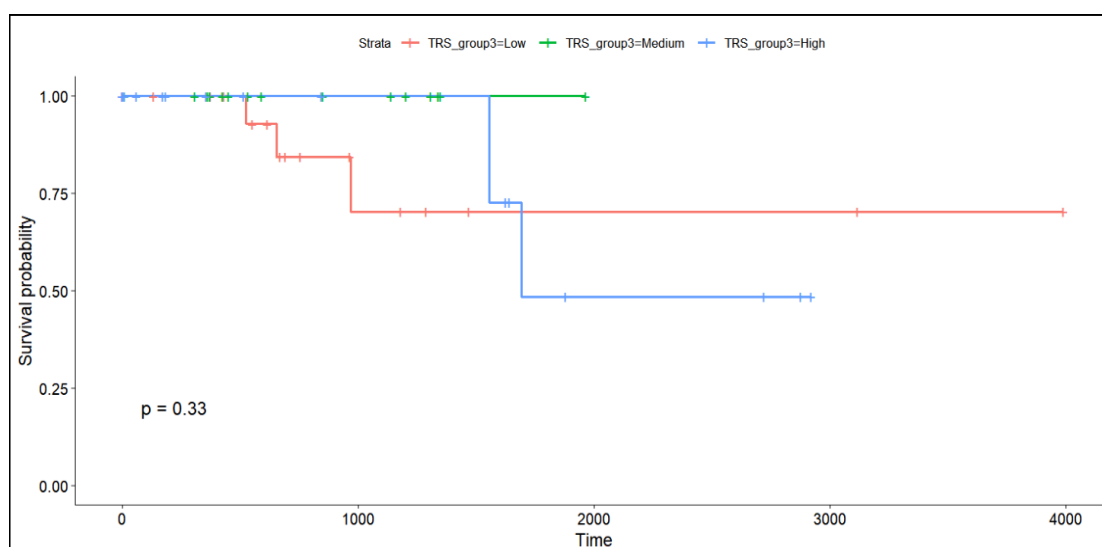


Fig 14: Kaplan–Meier Survival Analysis Stratified by Tumor Response Score (TRS) by Tertile Grouping (Low, Medium, High)

5.9. Spearman Correlation Heatmap of Key Immune Variables

A Spearman correlation matrix was computed across the four key analytical variables — CD274 expression, CD8⁺ T cell fraction, log-TMB, and TRS — and visualized as a heatmap with numerical correlation coefficients displayed within each cell.

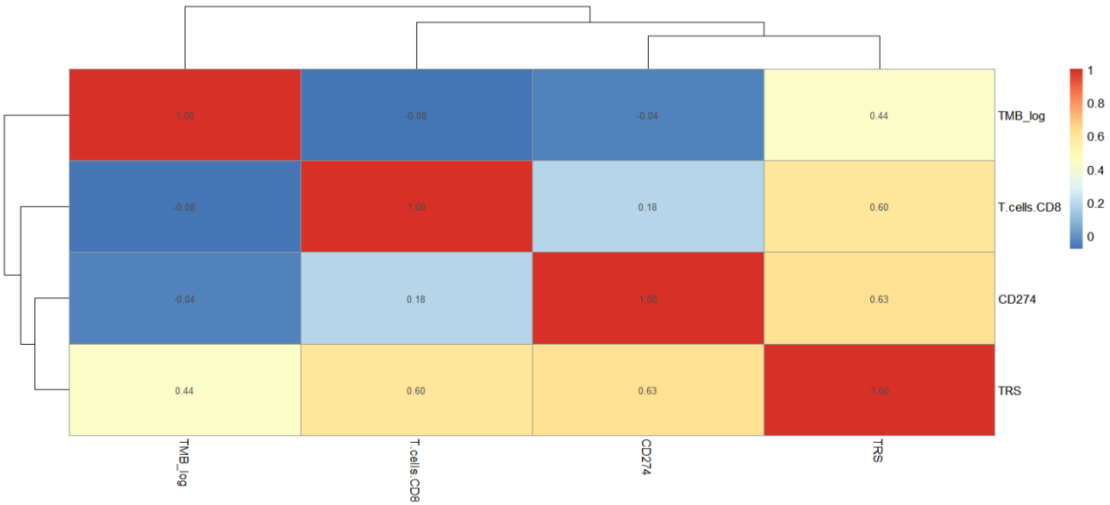


Fig 15: Spearman Correlation Heatmap of Key Immune and Molecular Variables (CD274, CD8⁺ T-cell Fraction, log2-TMB, and TRS)

5.10. IDO2 Expression Analysis

IDO2 expression was extracted from the TNBC expression matrix and merged with the integrated analysis dataframe. Samples were stratified into IDO2-high and IDO2-low groups using a median split, and Kaplan-Meier survival analysis was performed to assess the prognostic significance of IDO2 expression.

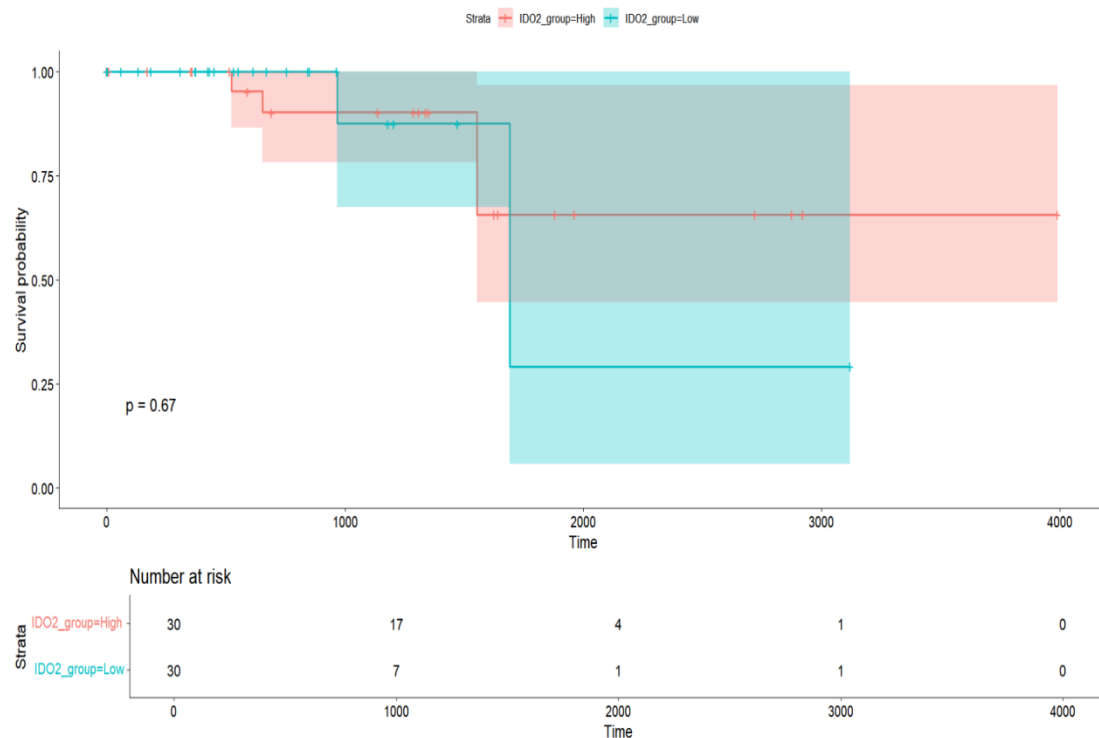


Fig 16: Kaplan–Meier Survival Analysis by IDO2 Expression (High vs. Low, Median Split) in TNBC

5.11. IDO2 and Immunosuppressive Cell Correlations

Scatter plots with linear regression overlays were generated for IDO2 versus Treg fractions and IDO2 versus M2 macrophage fractions. Spearman correlation results showed no statistically significant association between IDO2 and Tregs ($\rho = 0.054$, $p = 0.684$) or between IDO2 and M2 macrophages ($\rho = -0.167$, $p = 0.202$), suggesting that IDO2 upregulation in PD-L1-high TNBC may operate through pathways independent of these two immunosuppressive populations in this cohort.

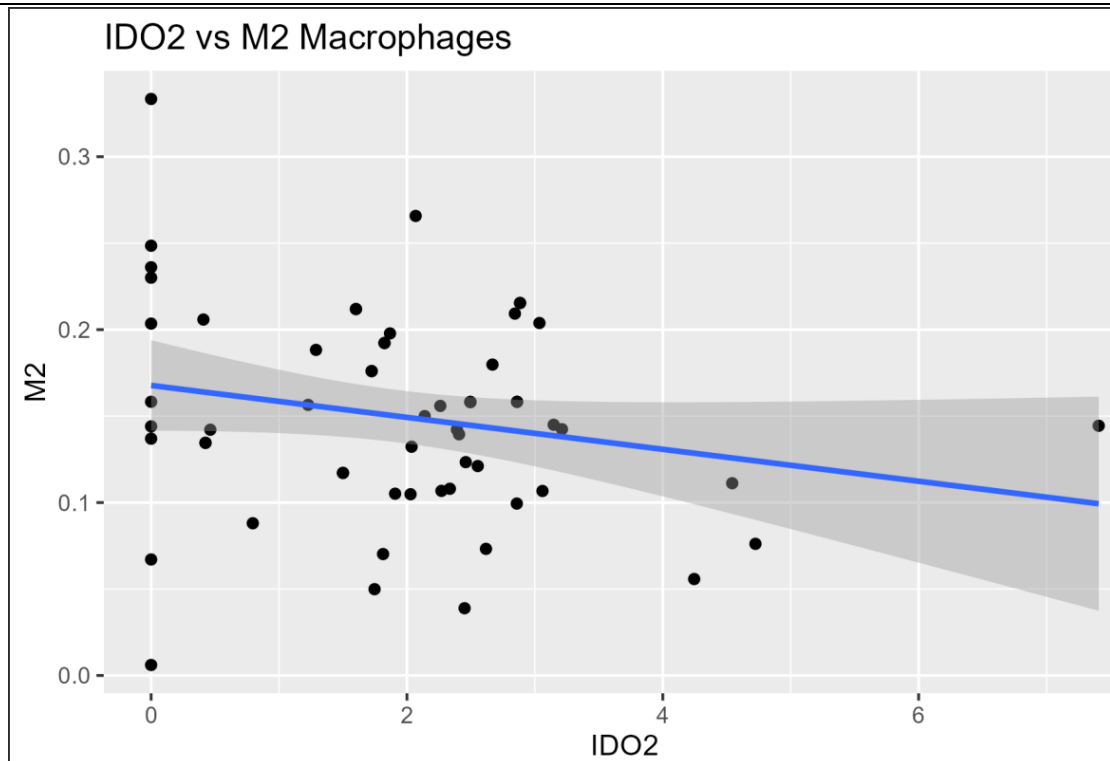


Fig 17: Scatter Plots of IDO2 Expression vs. Immunosuppressive Cell Populations (M2 Macrophages) using Spearman Correlation

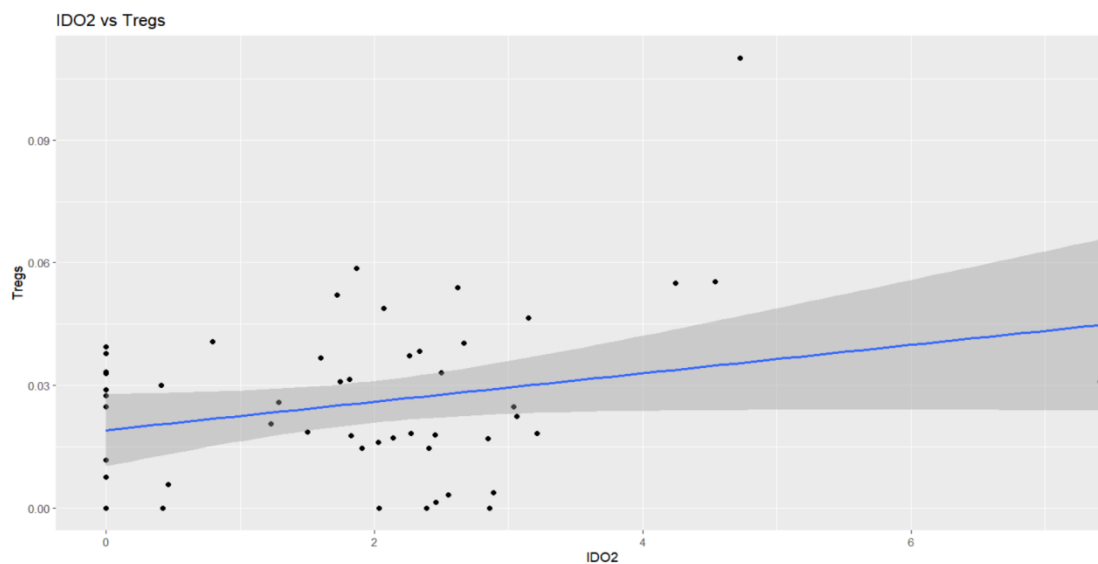


Fig 18: Scatter Plots of IDO2 Expression vs. Immunosuppressive Cell Populations (Tregs) using Spearman Correlation

5. 12. DISCUSSION:

This study employed an integrative transcriptomic approach to characterize the PD-L1-associated immune landscape in TNBC, combining differential gene expression, immune cell deconvolution, mutational burden analysis, and clinical survival data from the TCGA-BRCA cohort.

PCA of the 76 Basal-like TNBC samples revealed a discernible separation trend between PD-L1-high and PD-L1-low tumors along the first two principal components, indicating that PD-L1 expression is associated with broader transcriptomic reorganization rather than isolated gene-level changes. This is consistent with the established role of PD-L1 as a downstream effector of interferon-gamma signaling, which remodels gene expression programs across the tumor microenvironment.

DEG analysis identified a predominantly immune-activating signature in PD-L1-high tumors, with upregulation of genes including KLRC4, IFNG, IL21, NCR1, and IDO2 being associated with cytotoxic NK/T-cell activity and interferon response pathways. Notably, IDO2, a tryptophan-catabolizing enzyme implicated in immune tolerance, emerged as a significantly upregulated gene in PD-L1-high tumors. However, Spearman correlation analysis revealed no statistically significant association between IDO2 expression and either Treg fractions ($\rho = 0.054$, $p = 0.684$) or M2 macrophage fractions ($\rho = -0.167$, $p = 0.202$), suggesting that IDO2 upregulation in this context may reflect adaptive immune resistance rather than active immunosuppression via these two cell types. This is consistent with the broader literature indicating that IDO-mediated immune evasion is context-dependent and may involve alternative downstream effectors.

CIBERSORTx deconvolution confirmed that PD-L1-high tumors are enriched for activated CD8⁺ T cells, M1 macrophages, and dendritic cell subsets, while suppressive populations showed negative correlations with PD-L1, collectively painting a picture of an immunologically "hot" but simultaneously stressed tumor microenvironment. The strong positive association between PD-L1 and CD8⁺ T-cell infiltration aligns with the adaptive immune resistance model, wherein tumor cells upregulate PD-L1 in response to cytotoxic T-cell pressure.

TMB distribution across the BRCA cohort was predominantly low (0–5 mut/Mb), consistent with published reports characterizing breast cancer as a relatively hypomutated tumor type compared to melanoma or lung cancer. The composite TRS, integrating PD-L1, CD8⁺ T-cell fraction, and log-TMB, did not yield statistically significant survival stratification (binary: $p = 0.97$; tertile: $p = 0.33$). This is likely attributable to the modest cohort size of 76 samples and the retrospective nature of TCGA data, which limits statistical power for survival endpoints. Nonetheless, the TRS framework represents a methodologically meaningful approach toward multi-dimensional biomarker integration, and its validation in larger prospective cohorts or clinical trial datasets remains a compelling future direction.

Taken together, these findings underscore the inadequacy of single-parameter biomarkers and support the growing consensus that composite immune profiling is necessary to capture the biological complexity of the TNBC tumor microenvironment.

6. CONCLUSION:

This multi-layered analytical approach captures the transcriptomic, immune, and mutational dimensions of TNBC biology in an integrated framework.

PD-L1-high tumors were found to harbor a distinct immune-activating transcriptomic signature, marked by upregulation of cytotoxic and interferon-response genes including KLRC4, IFNG, IL21, and IDO2, alongside enrichment of activated CD8⁺ T cells and M1 macrophages. These findings are consistent with the adaptive immune resistance model and reinforce the biological rationale for PD-1/PD-L1 checkpoint blockade in immunologically active TNBC subgroups.

The absence of significant correlations between IDO2 and immunosuppressive populations such as Tregs and M2 macrophages suggests that IDO2-mediated immune evasion in TNBC may operate through alternative, yet unexplored pathways. The composite Tumor-Reactive Score (TRS), while not yielding statistically significant survival stratification in this cohort, demonstrates the feasibility and conceptual value of integrating orthogonal immune parameters into a single metric. Taken together, these results underscore the inadequacy of single biomarkers and support the development of composite, data-driven frameworks for

guiding immunotherapy decisions in TNBC. Validation of these findings in larger, prospective cohorts remains an important next step.

7. FUTURE DIRECTIONS:

1. **Expand Immune-Checkpoint Panel** – Incorporate PD-1, CTLA-4, LAG3, VISTA, TIGIT to identify multi-checkpoint tumor subgroups.
2. **Multi-Omics Integration** – Combine transcriptomic, mutational, immune-cell, and clinical data to build a composite **immune activation score**.
3. **Microenvironment Analysis** – Explore spatial relationships and classify tumors as immune-hot, immune-cold, or immune-excluded.
4. **Predictive Modeling** – Develop and validate a machine-learning–based composite biomarker for PD-1/PD-L1 therapy response, comparing performance to PD-L1 alone.

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ANNEXURE A: DEG Table:

	logFC	AveExpr	t	P.Value	adj.P.Val	B	Significance
KLRC4	1.7293465	3.3604862	8.8624871	2.471661853	1.691440594	17.098795	Up
	3240149	7748993	3928994	34941e-13	97544e-09	6396366	
CLEC6A	1.4103835	3.7947611	7.5420878	8.298480777	1.002163590	12.703607	Up
	2128576	3692049	8964845	94321e-11	41867e-07	2688612	
IL21	1.1844992	2.7024632	6.9176156	1.261021650	6.020645226	10.230420	Up
	4274584	3877441	5004075	07144e-09	969e-07	3958422	
IFNG	1.2470905	4.4944536	6.7236896	2.911095761	1.078004188	10.181707	Up
	0341838	1814097	8752955	57789e-09	46606e-06	1189683	
SSTR3	1.2042336	3.4561316	6.7175900	2.988476336	1.078004188	9.5623733	Up
	547475	6436815	8172777	86926e-09	46606e-06	2179943	
IDO2	1.2102130	3.8803636	6.5689834	5.653511017	1.546380758	9.2378529	Up
	805229	7448281	0113003	34584e-09	75136e-06	3492748	
CXCR2P1	1.0231826	5.1885927	6.5660651	5.724546403	1.546380758	10.103136	Up
	5682858	0181805	299745	56081e-09	75136e-06	2666614	
NCR1	1.1781545	3.4250663	6.4964266	7.708599559	1.872445355	8.7846711	Up
	3297052	5789503	9916787	81726e-09	43022e-06	4628004	
LOC400696	1.3596472	4.0889675	6.3903289	1.211204502	2.590211296	8.7701511	Up
	9028574	8924347	8523755	75815e-08	00259e-06	3509629	
IL12B	1.1171549	4.1880929	6.2238869	2.451129862	4.124729187	8.1693402	Up
	8667294	7893791	2942305	95089e-08	40834e-06	7428166	
KIR3DL1	1.2325527	3.4796927	6.0615050	4.851246950	6.552374992	7.3035148	Up
	7860587	0508098	4033919	23765e-08	65651e-06	5680142	
CR1L	1.1941838	3.8313408	5.9989966	6.300325508	8.134948596	7.2106644	Up
	2882262	6924575	8011241	04186e-08	23266e-06	6790202	
CLEC4C	1.2943599	3.3421656	5.9871343	6.620078115	8.393145209	7.0266704	Up
	7135679	7473215	7078537	97348e-08	01172e-06	083441	
KIR2DL4	1.1229091	4.2607070	5.7189554	2.010209199	1.905813503	6.4204553	Up
	8910804	34	3574265	32679e-07	07167e-05	3658138	

KLRC3	1.1569548 0830135	4.0014870 358016	5.6994180 1071743	2.178175500 40021e-07	2.023436335 89214e-05	6.2462122 2068798	Up
CASP 5	1.0209735 2346506	3.8406495 5549758	5.6882712 116077	2.280123518 94716e-07	2.080486037 51045e-05	6.1194405 9566886	Up
CLNK	1.0007911 9411148	3.1981848 3601748	5.6805791 5281793	2.353200960 95209e-07	2.128247388 91394e-05	5.9934591 4285117	Up
KIR2D L3	1.1271372 7454905	3.4217750 0964825	5.4240966 4408957	6.677553718 16822e-07	4.584955780 40112e-05	5.1402486 353137	Up
GBP7	1.0548621 6269194	3.0730596 8666393	5.2363033 3695708	1.416287778 49907e-06	7.995585423 92836e-05	4.5215841 2567616	Up
KLRC2	1.0399222 0407779	4.1894476 1617126	5.1042003 4473587	2.387979154 86688e-06	0.000117849 067426483	4.2766235 7292139	Up
KIR3D L2	1.0769325 0320035	3.7428945 7905013	4.8770310 512388	5.784937785 97701e-06	0.000227955 417938787	3.4150983 3901505	Up
CNR2	1.1033026 9982788	3.8001164 9134351	4.6974653 7905083	1.149182088 61251e-05	0.000379303 991627247	2.8597287 9596013	Up
CD30 OE	1.0301594 4449263	3.9479864 3540786	4.6095710 522986	1.600889940 20185e-05	0.000489810 290198868	2.6097694 4522292	Up
CLEC1 7A	1.0189372 4340356	3.9143665 2276895	4.3965486 8960351	3.529368675 55124e-05	0.000894542 455667494	1.9417965 1669427	Up
COL2 9A1	1.0163353 8568912	3.9315790 9136656	4.3673217 7006639	3.927899650 20558e-05	0.000968064 583658109	1.8555466 4853626	Up
FCRL1	1.0228842 4421379	4.0814175 4060408	4.1234666 8023771	9.450282657 72726e-05	0.001911470 96515409	1.1459836 6655978	Up
PSAPL 1	1.0439696 3572143	3.4753306 4371213	3.2605228 5263154	0.001666408 10950445	0.016997262 5474789	- 1.2616014 88	Up

Fig 19: Table of upregulated genes.

	logFC	AveExpr	t	P.Value	adj.P.Val	B	Significance
CCDC108	- 1.07010 6948	3.8586801 6929501	- 5.24786 2575	1.352650336 86461e-06	7.735351369 31208e-05	4.64844086 917612	Down
ADAMTS19	- 1.38227 4976	3.8279271 3830142	- 5.11689 9593	2.271569534 48182e-06	0.000113744 689129053	4.24656324 426153	Down
GDF1	- 1.09722 0348	4.0277267 2442035	- 4.15394 7535	8.480585445 30639e-05	0.001760428 90993064	1.23255128 295937	Down
ARHGDI	- 1.24205 0622	3.7376273 9190285	- 4.14102 126	8.879498410 98255e-05	0.001833964 81265062	1.15955200 483164	Down
TCN1	- 1.16856 9305	4.4766251 4067176	- 3.97639 4735	0.000158356 963144319	0.002882152 88417808	0.75967438 0227565	Down
CRISP2	- 1.01252 5256	3.2869233 0347201	- 3.63812 5391	0.000498384 405046268	0.006936835 14277958	- 0.28413002 8	Down
LOC100271831	- 1.02913 2901	3.6788481 0460261	- 3.11597 6162	0.002588001 94891912	0.023110778 6043104	- 1.61336723 7	Down

Fig 21: Table of downregulated genes

ANNEXURE B: CIBERSORTX IMMUNE CELL BOXPLOTS:

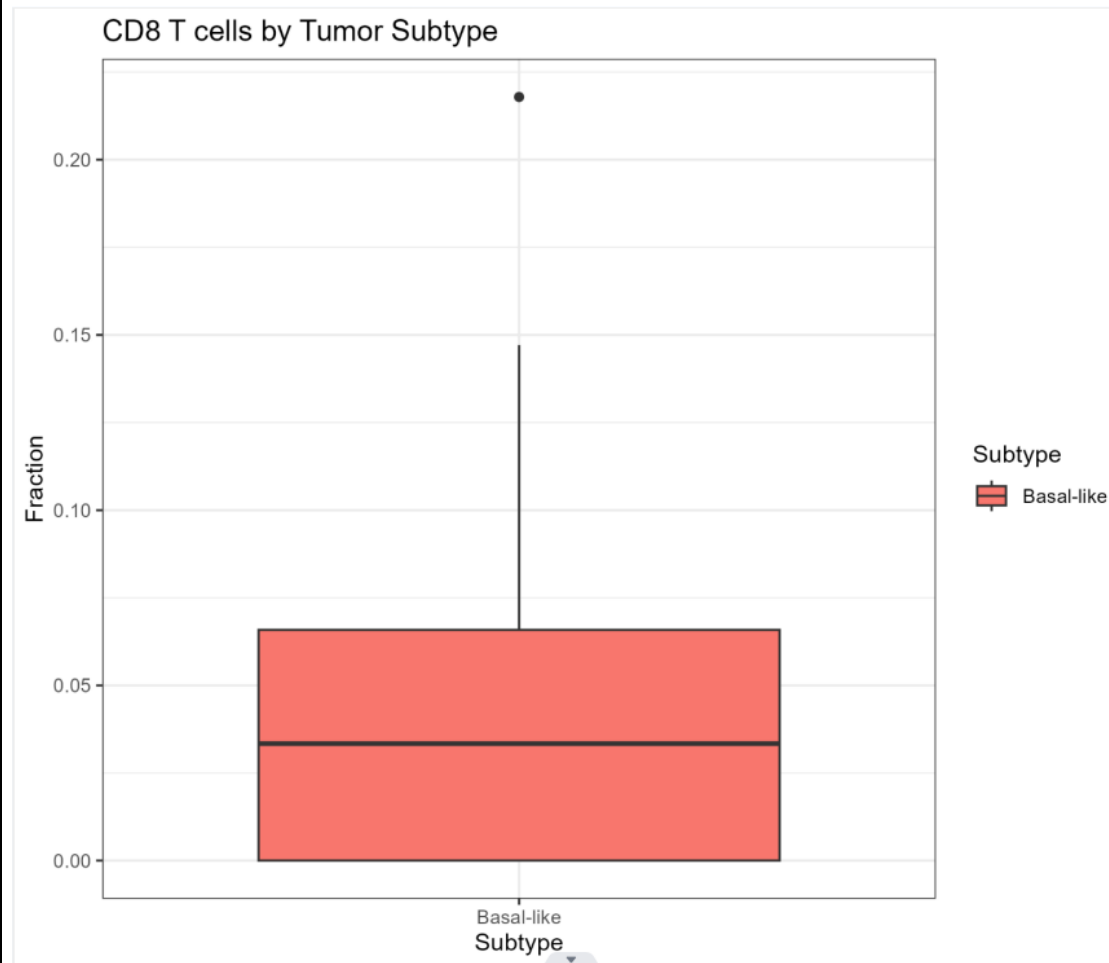


Fig 21: CD8 Tcells by Tumor subtype

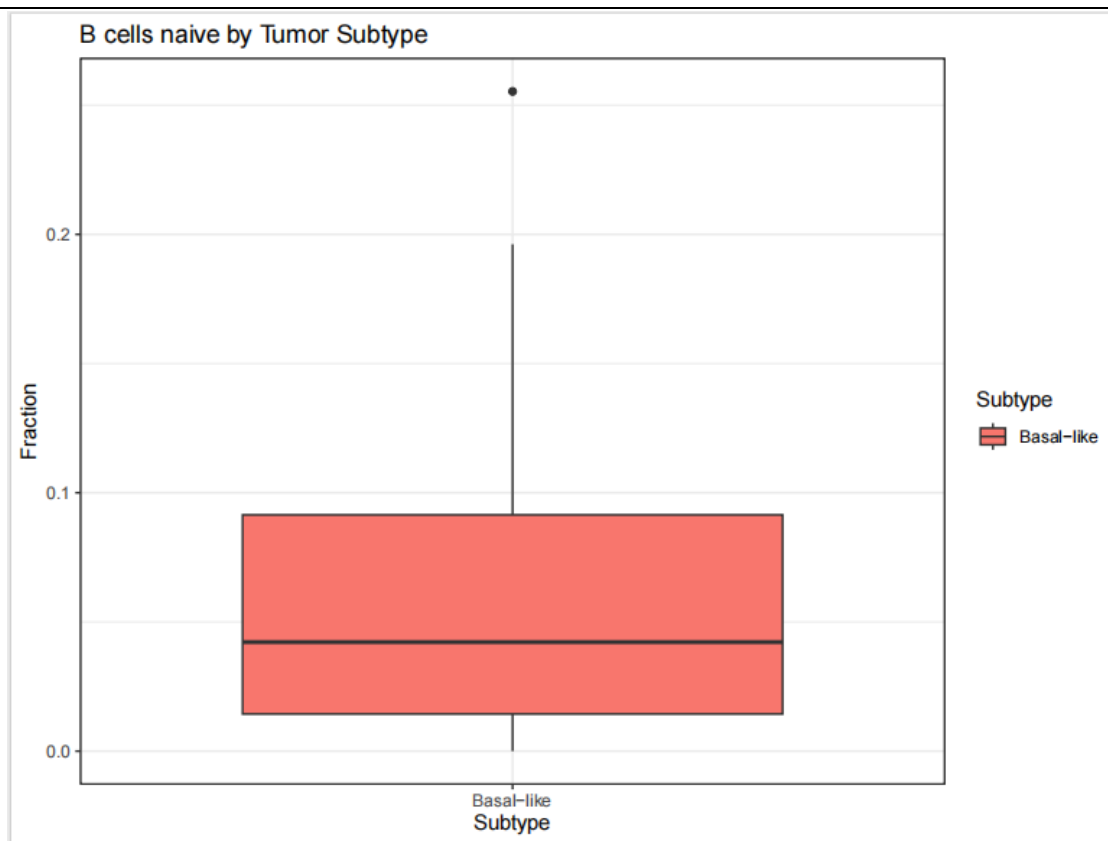


Fig 22: B cells naive by Tumor subtype

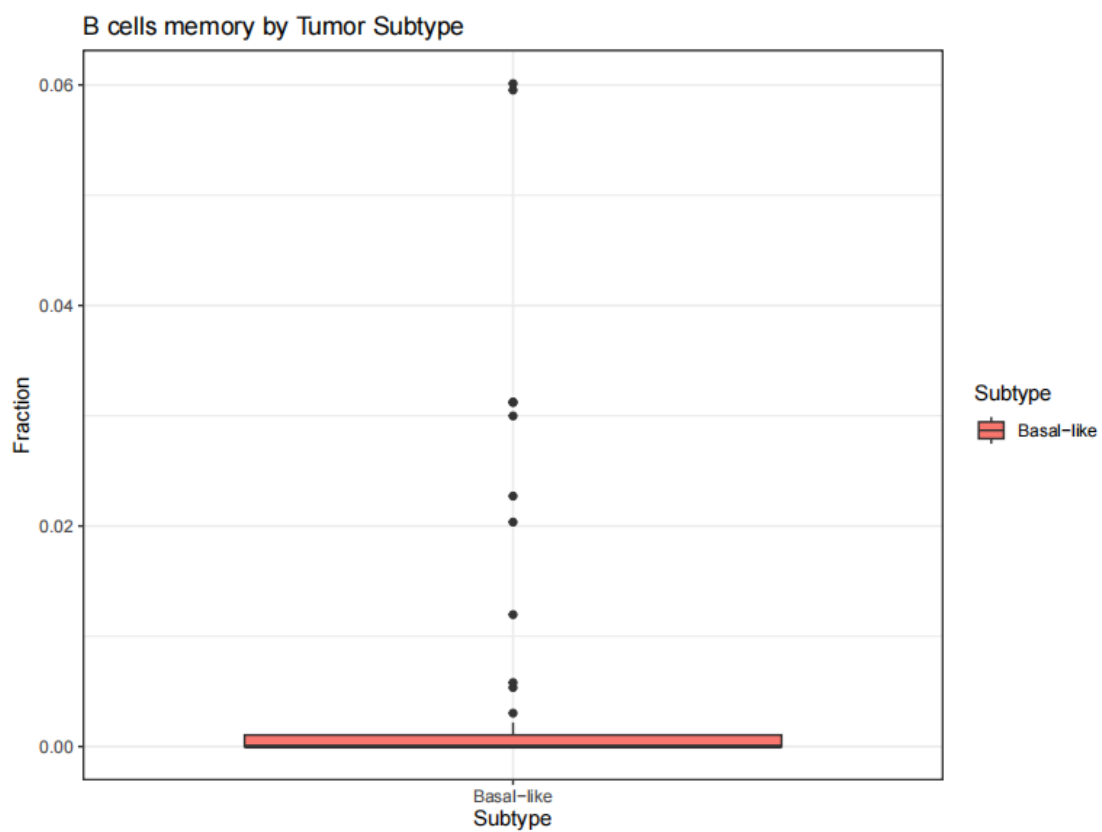


Fig 23: B cells memory by tumor Subtype

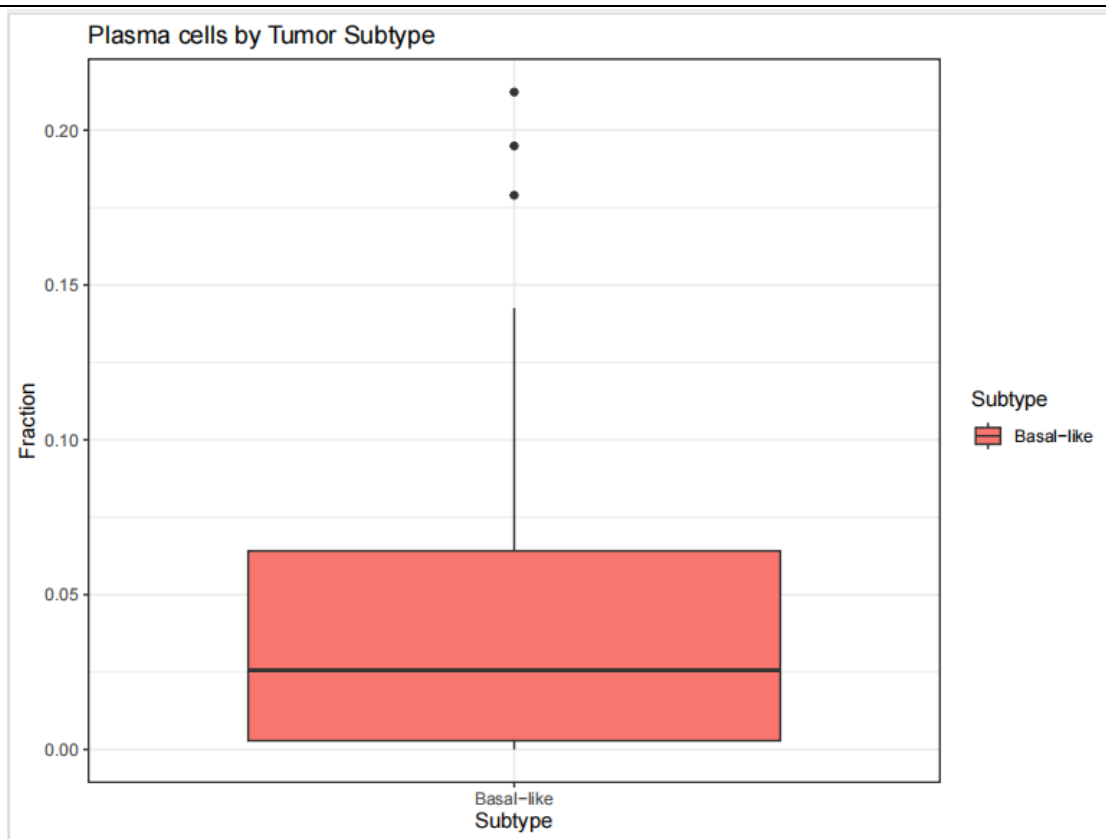


Fig 24: Plasma cells by tumor Subtype

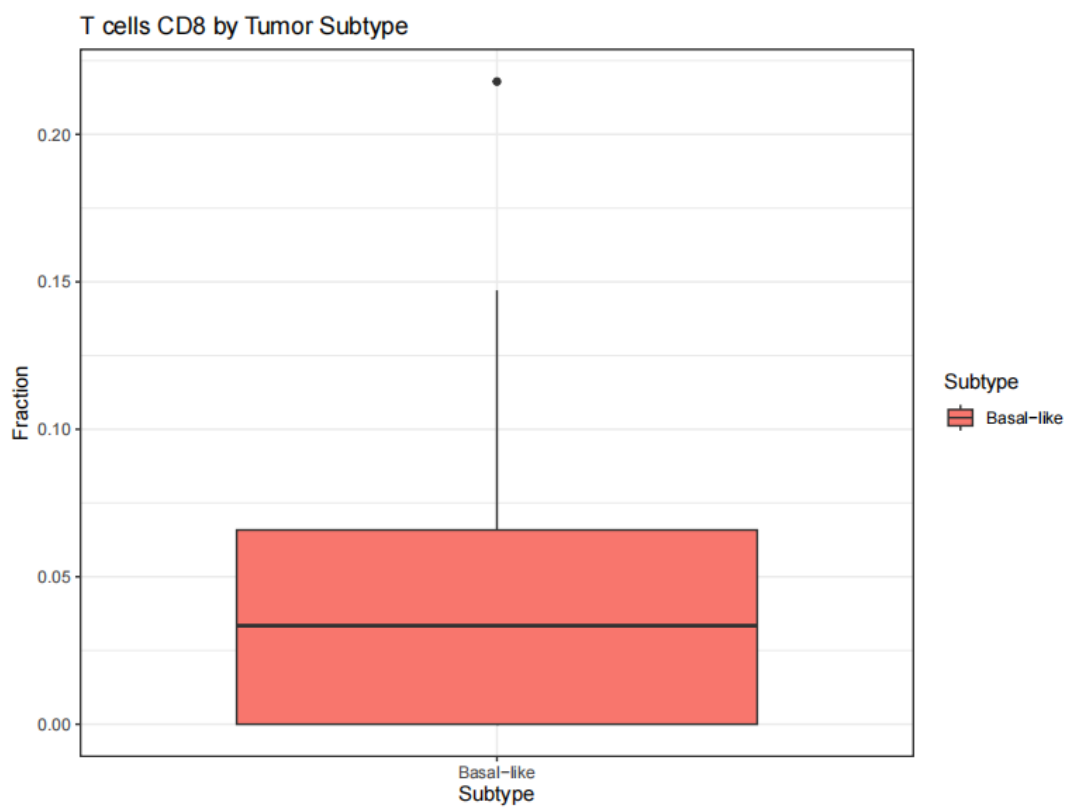


Fig 25: T cells CD8 by Tumor Subtype

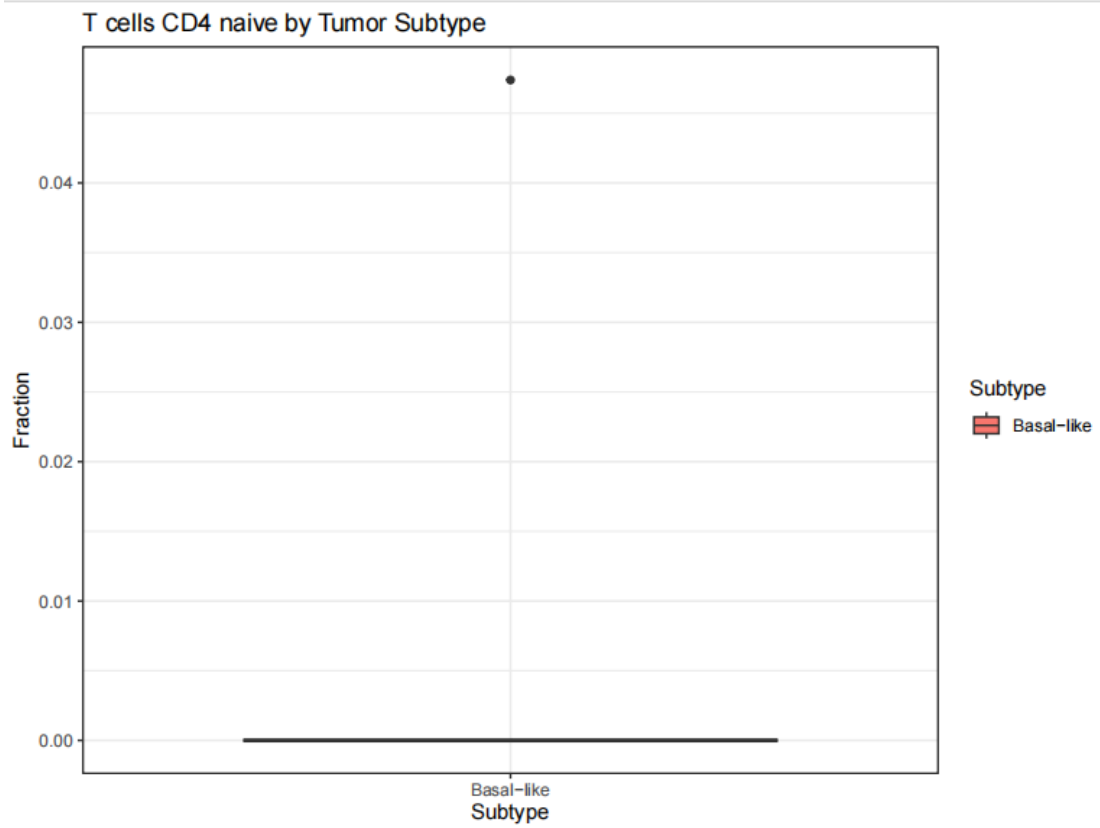


Fig 26: T cells CD4 naive by Tumor Subtype

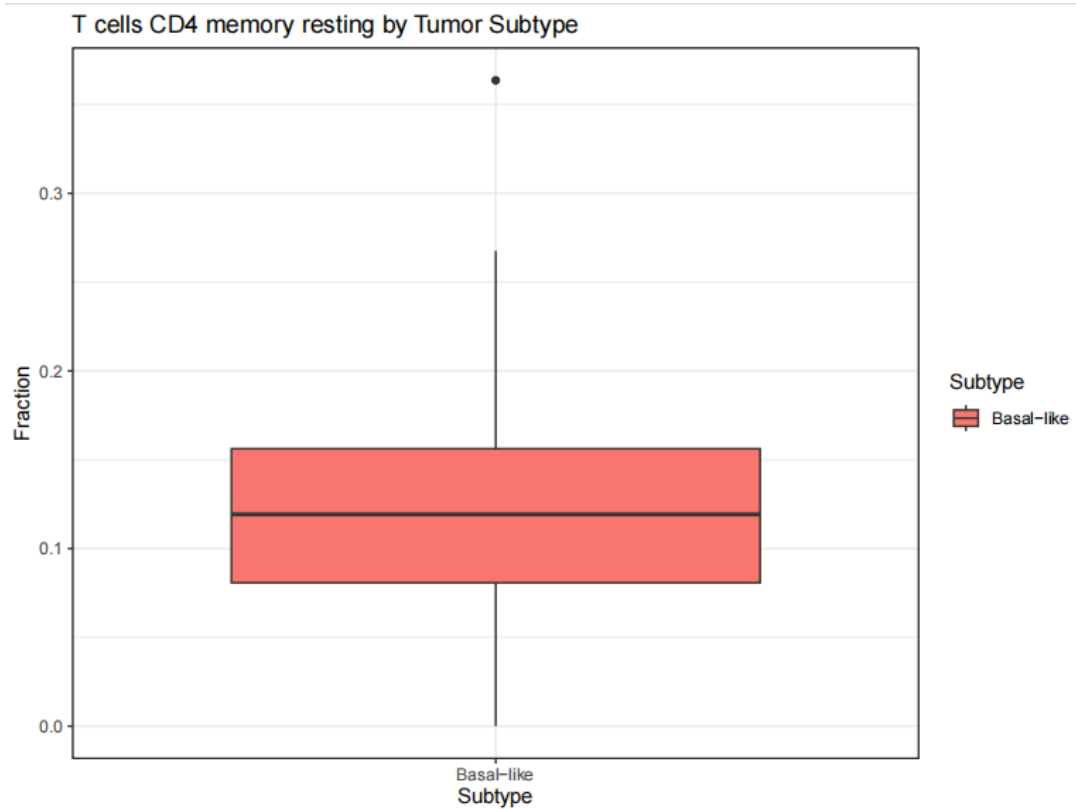


Fig 27: T cells CD4 memory resting by Tumor Subtype

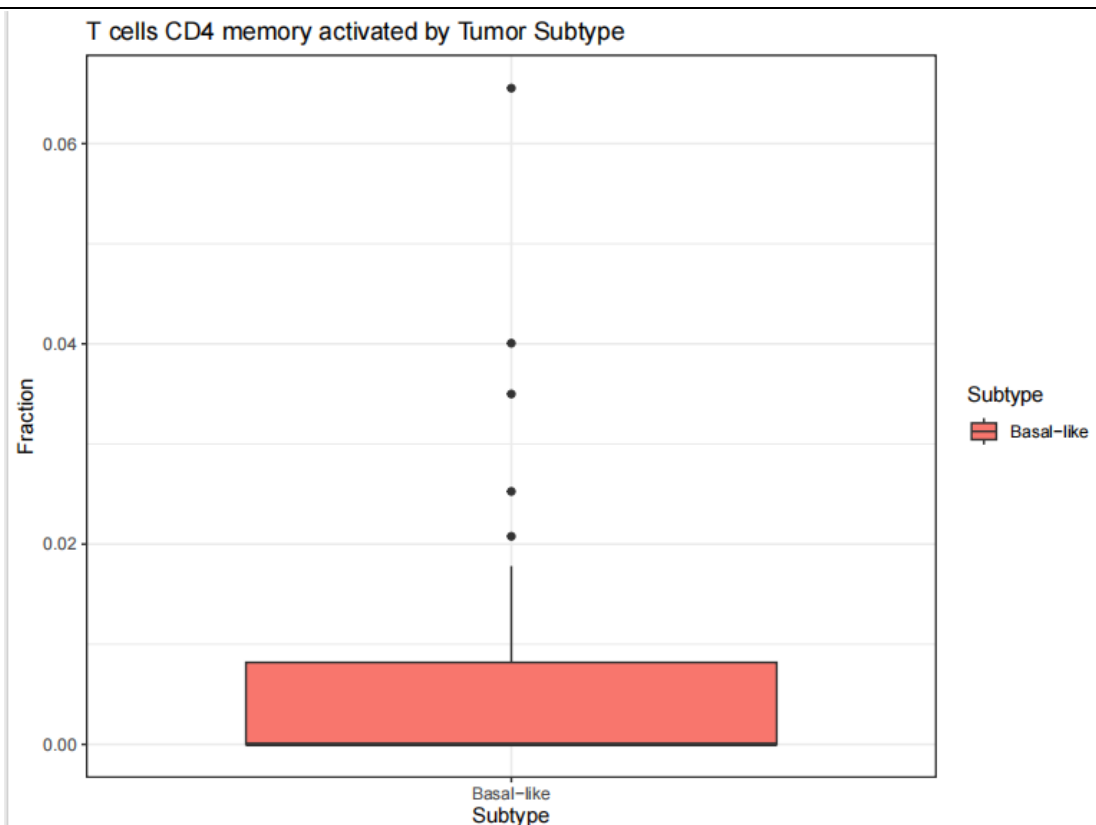


Fig 28: T cells CD4 memory activated by Tumor Subtype

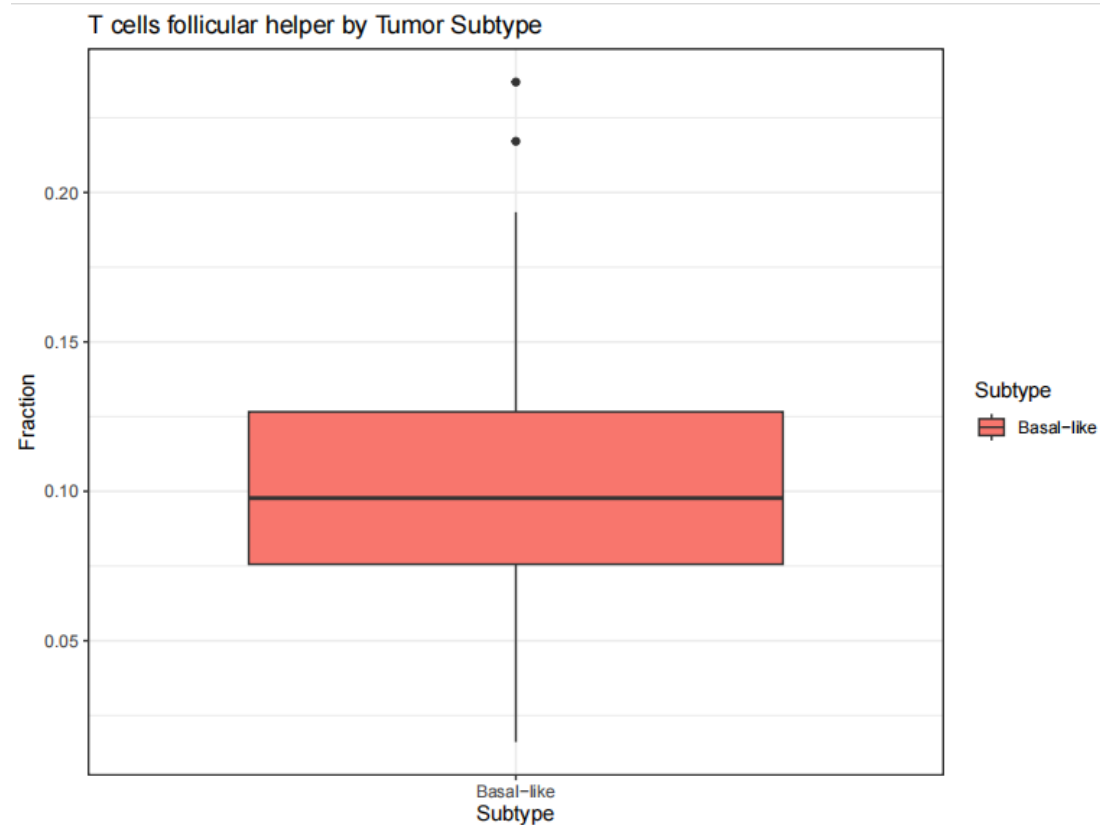


Fig 29: T cells follicular helper by Tumor Subtype

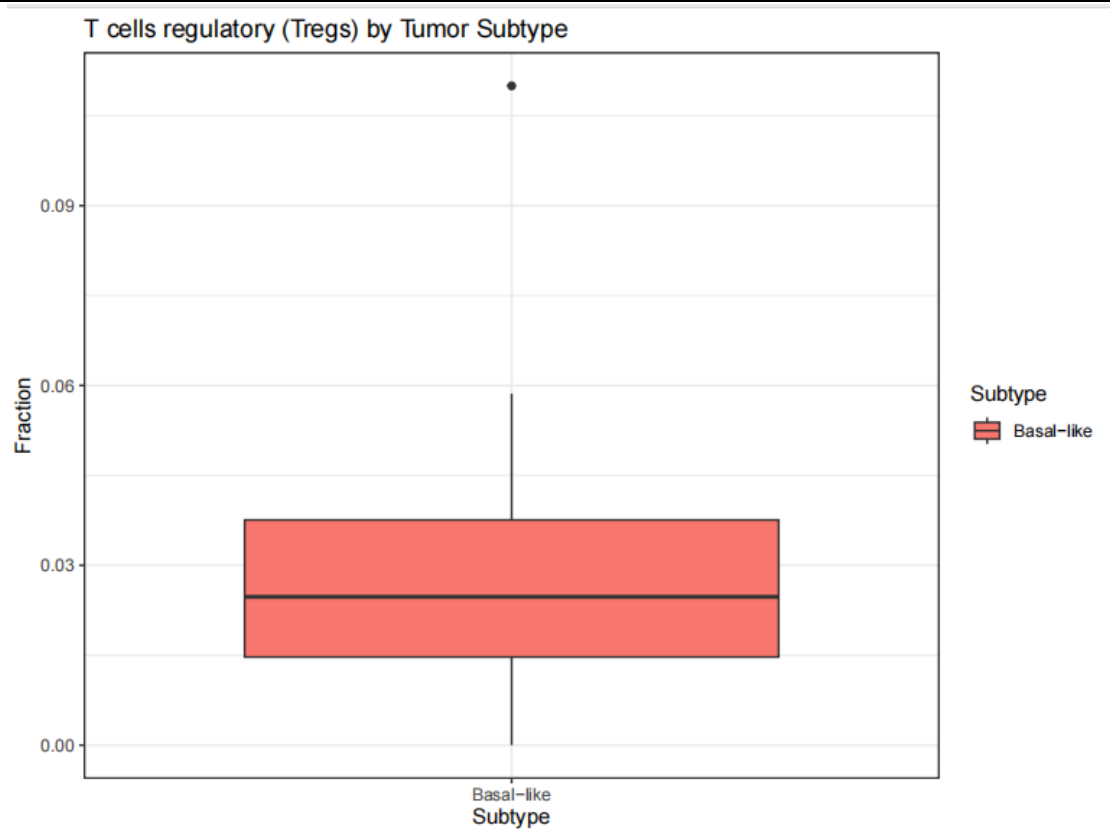


Fig 30: T cells regulatory (Tregs) by Tumor Subtype

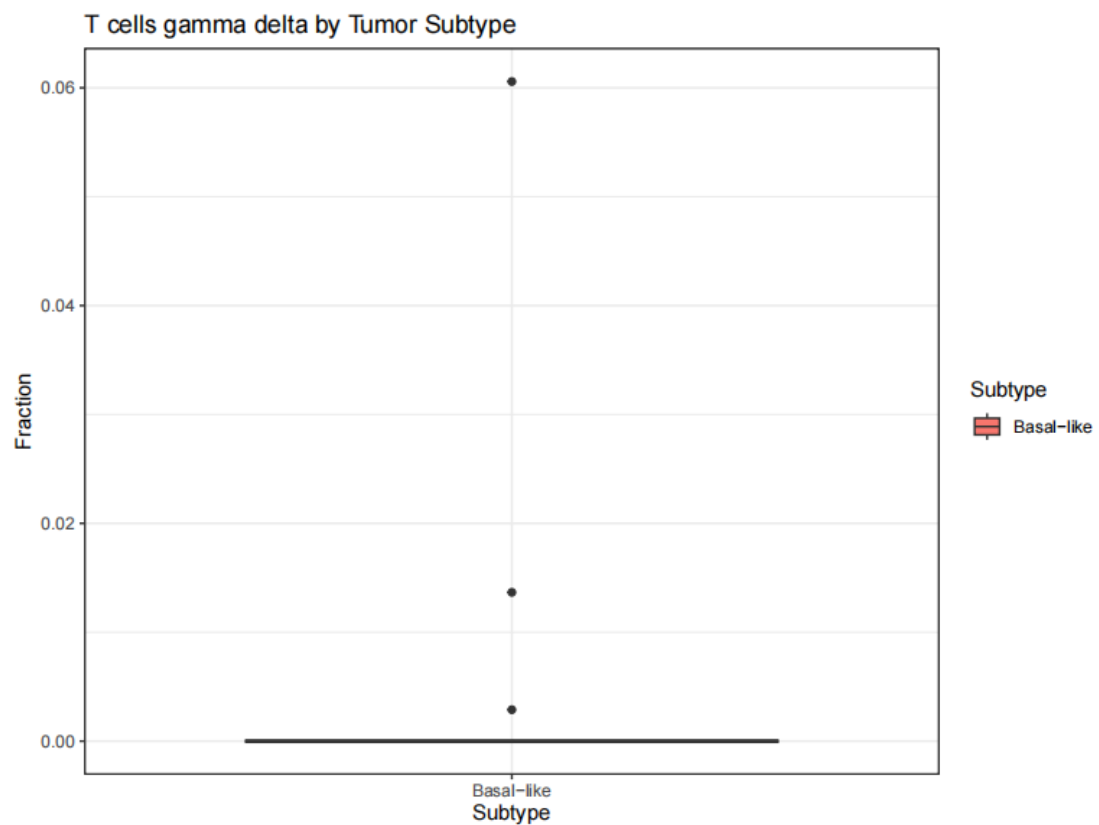


Fig 31: T cells gamma delta by Tumor Subtype

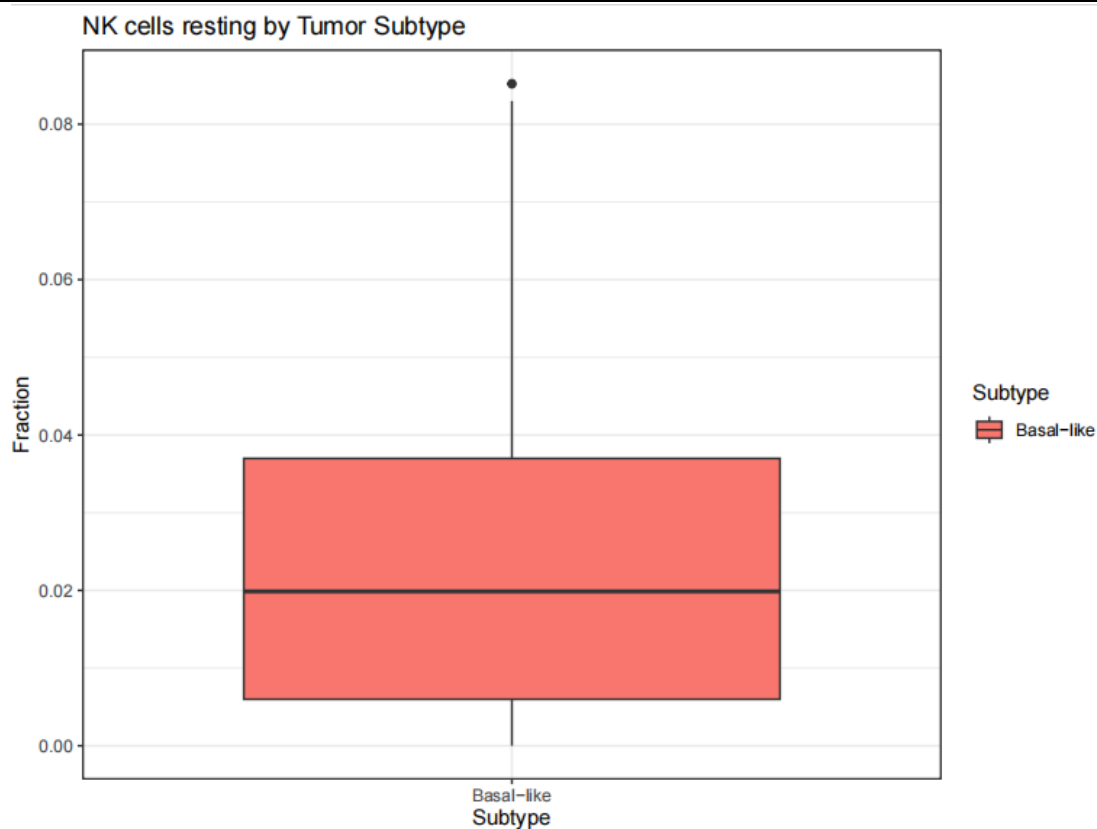


Fig 32: NK cells resting by Tumor Subtype

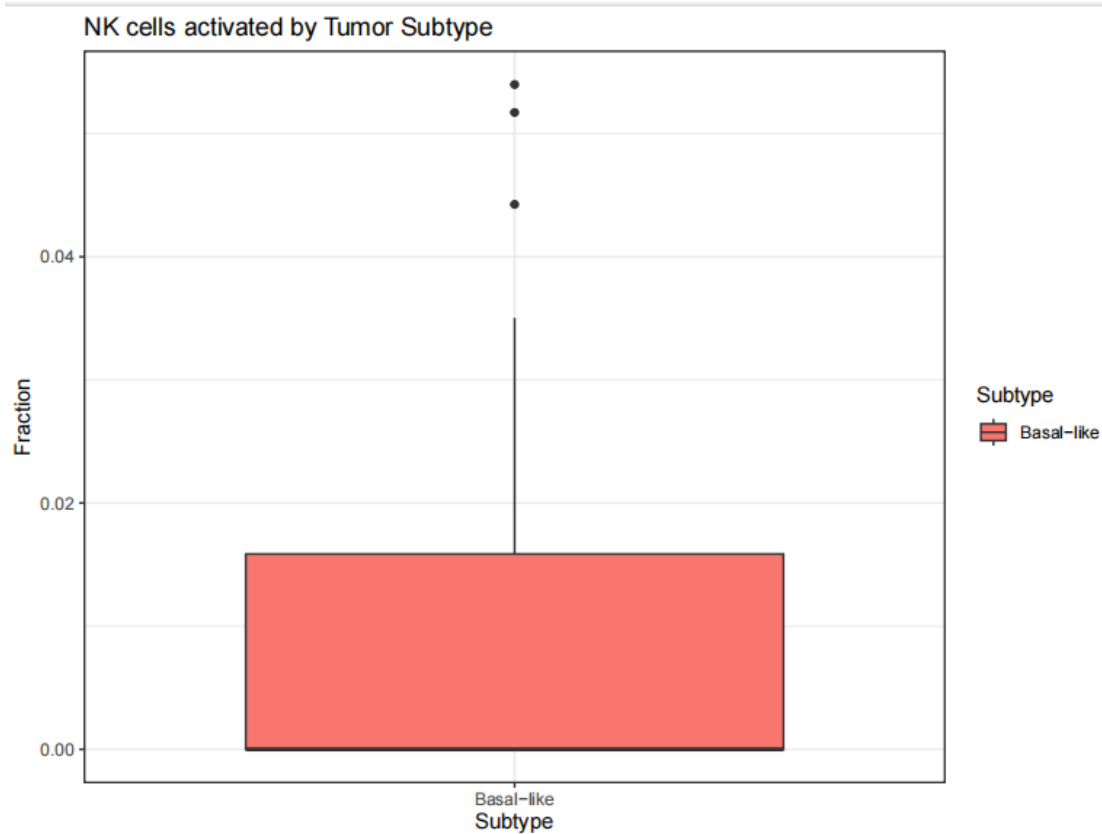


Fig 33: NK cells activated by Tumor Subtype

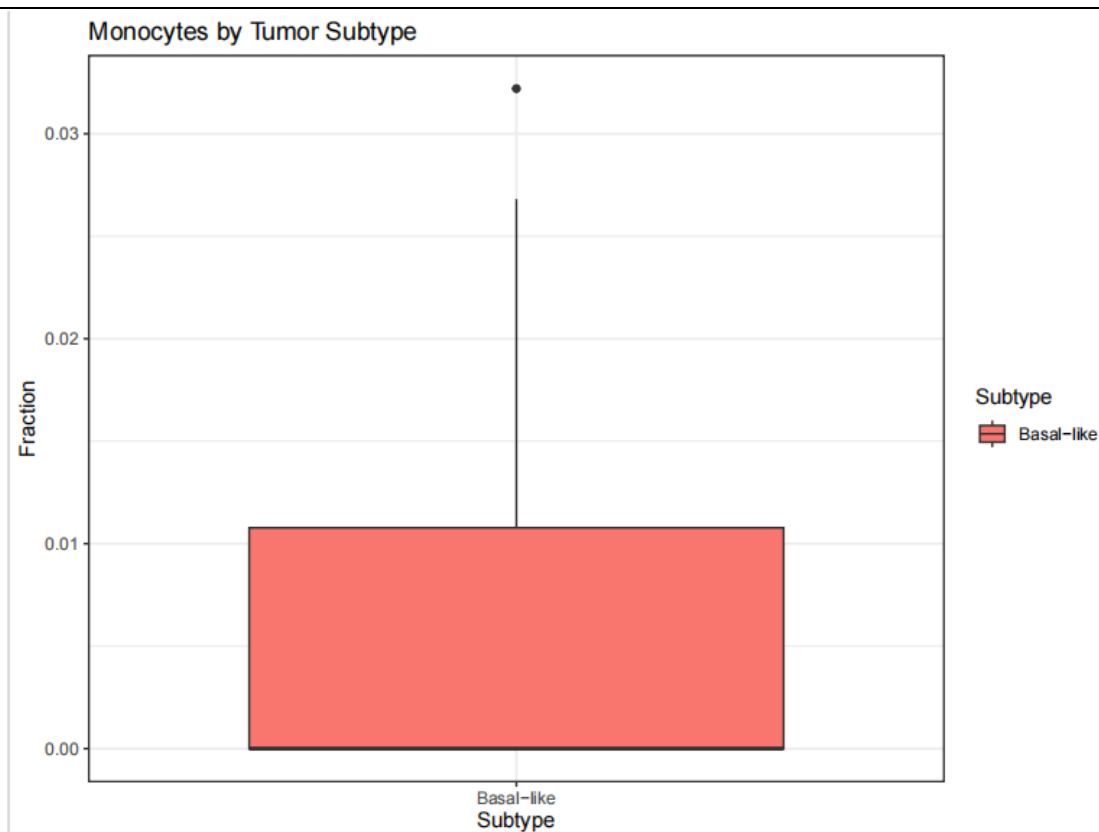


Fig 34: Monocytes by Tumor Subtype

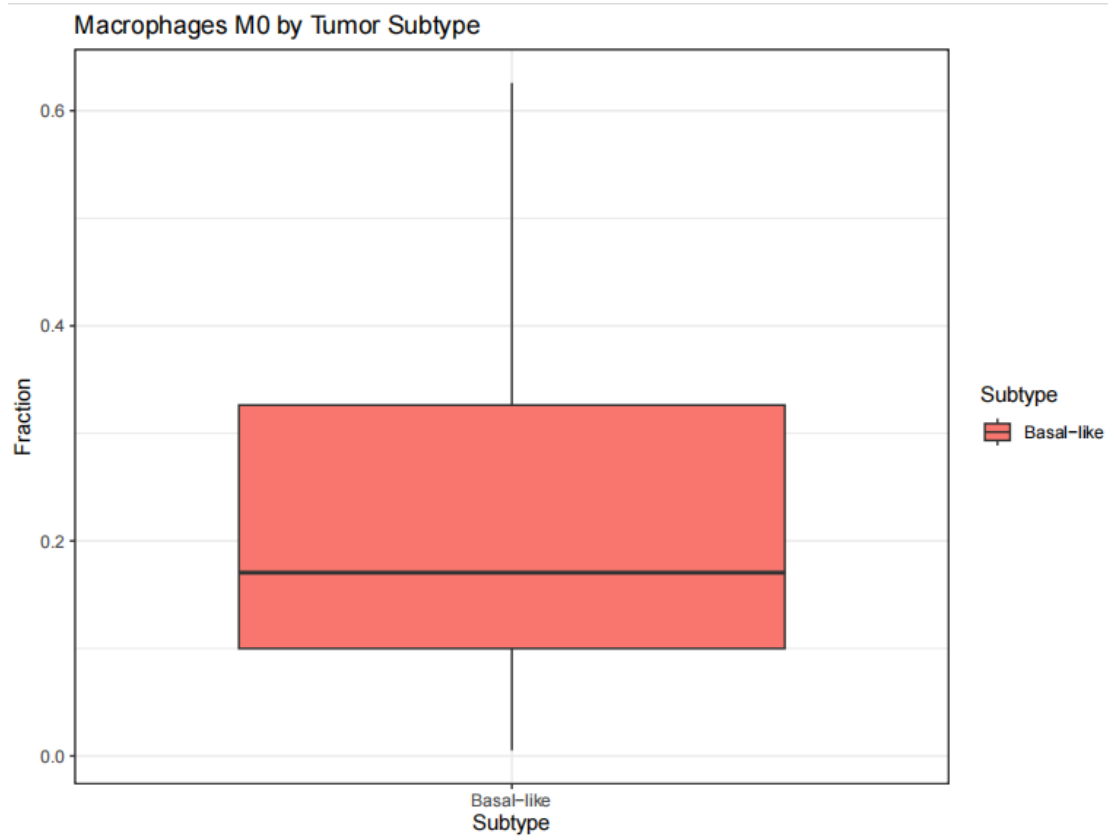


Fig 35: Macrophages M0 by Tumor Subtype

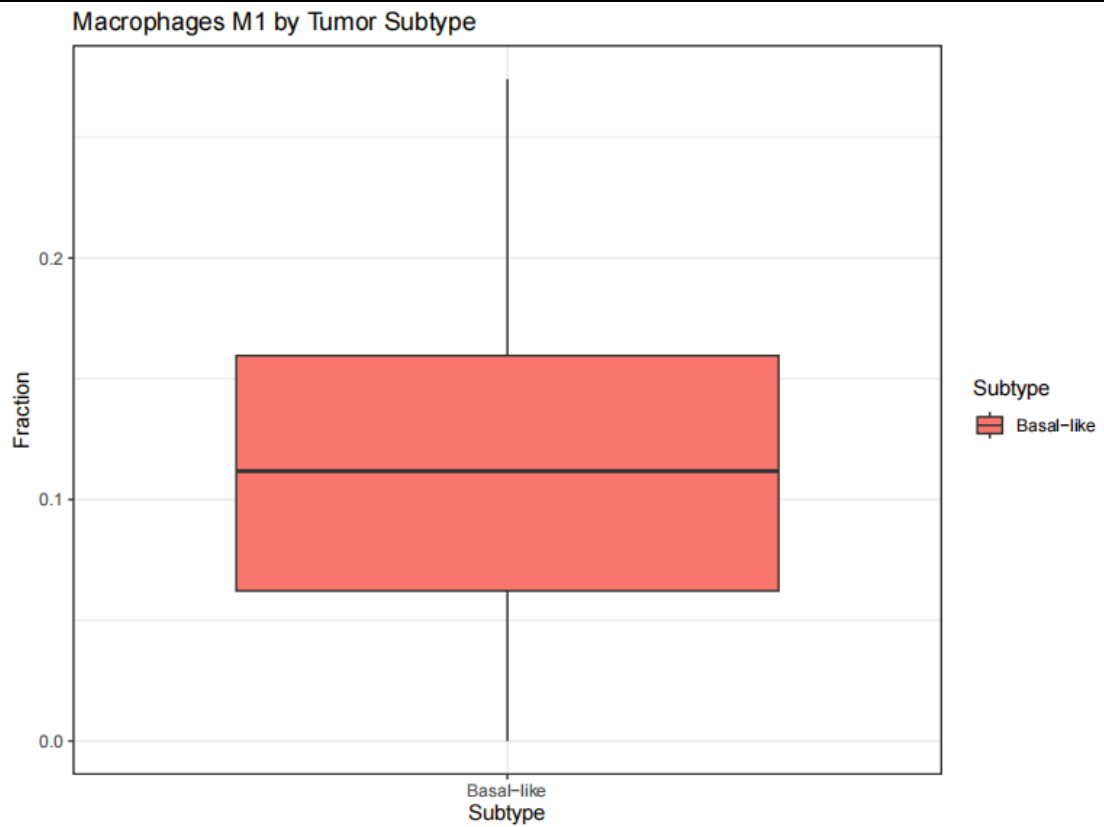


Fig 36: Macrophages M1 by Tumor Subtype

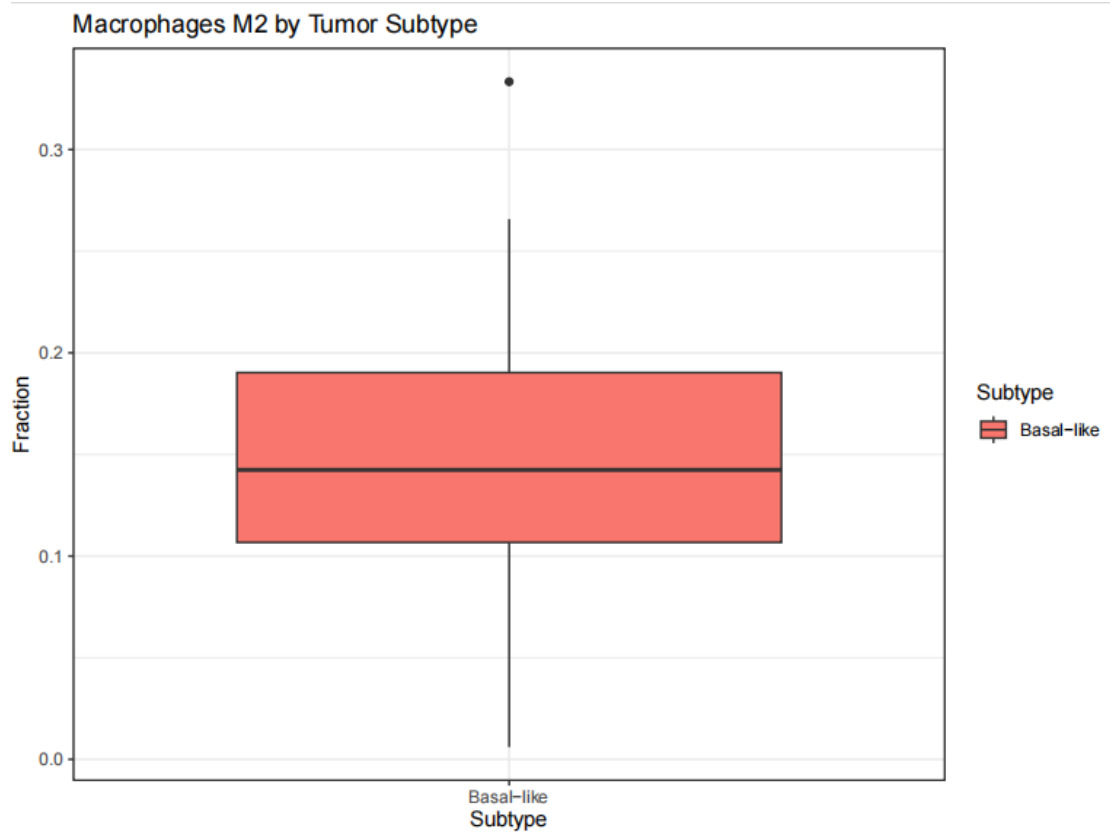


Fig 37: Macrophages M2 by Tumor Subtype

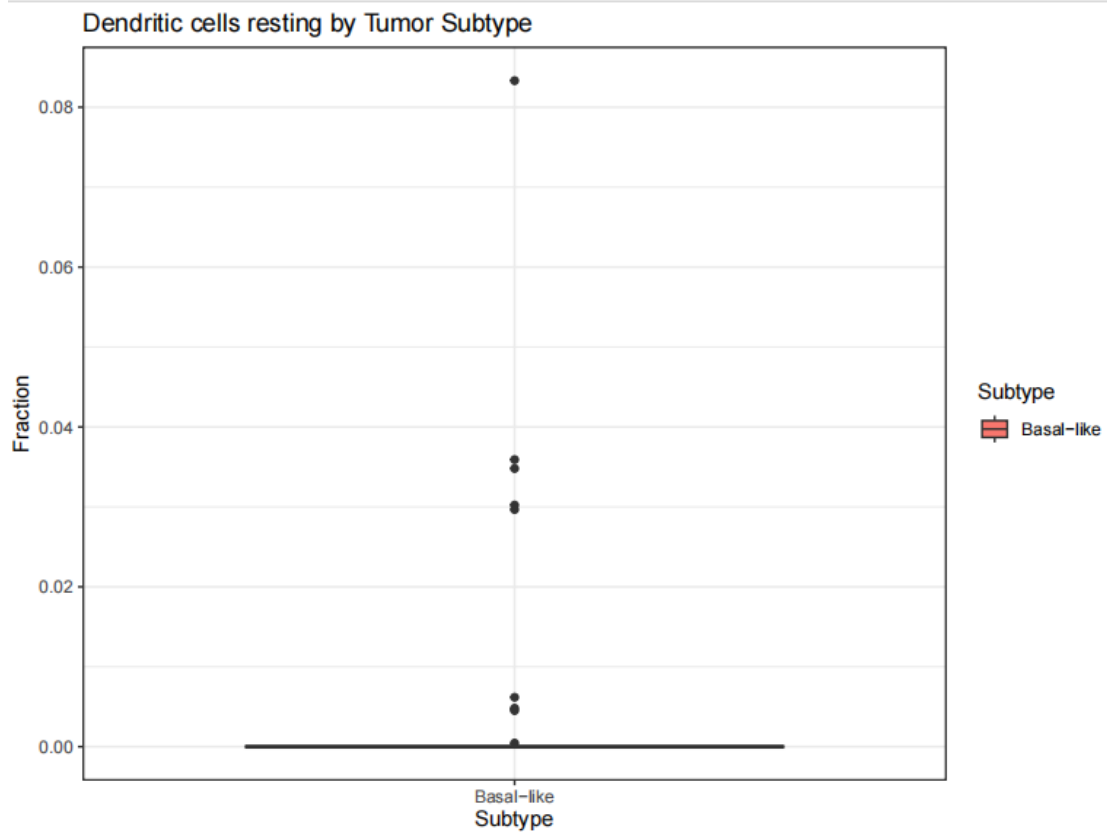


Fig 38: Dendritic cells resting by Tumor Subtype

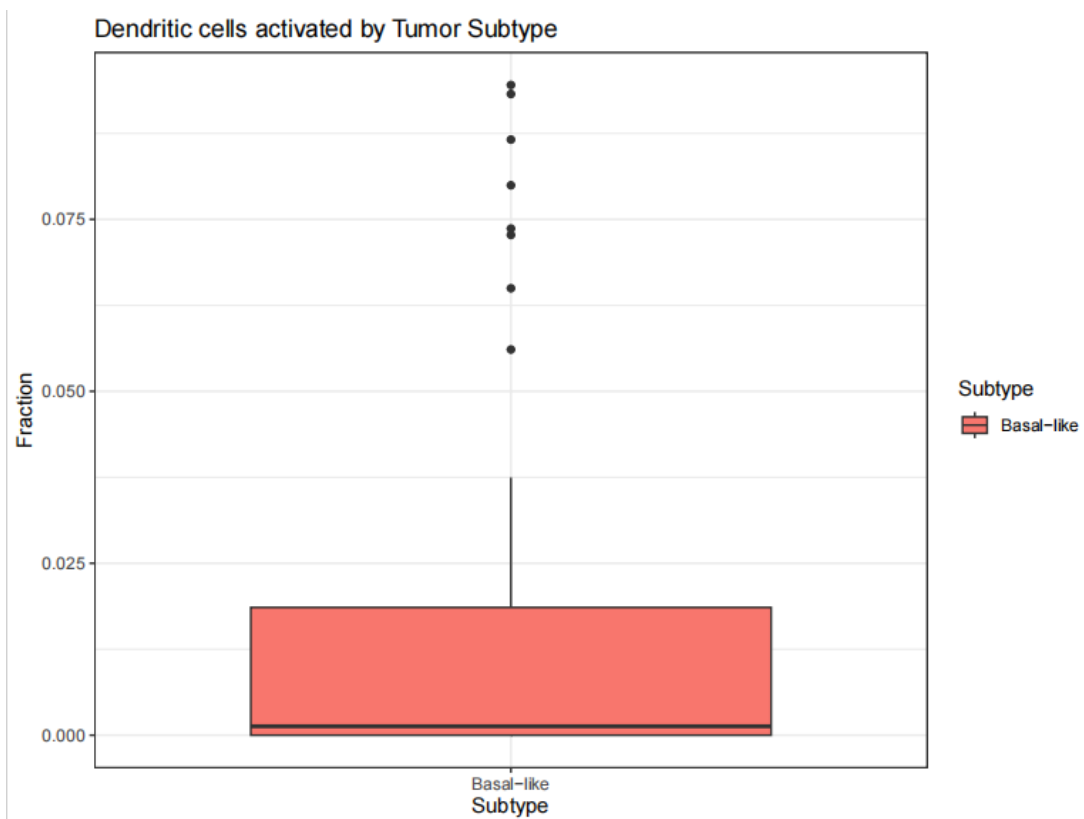


Fig 39: Dendritic cells activated by Tumor Subtype

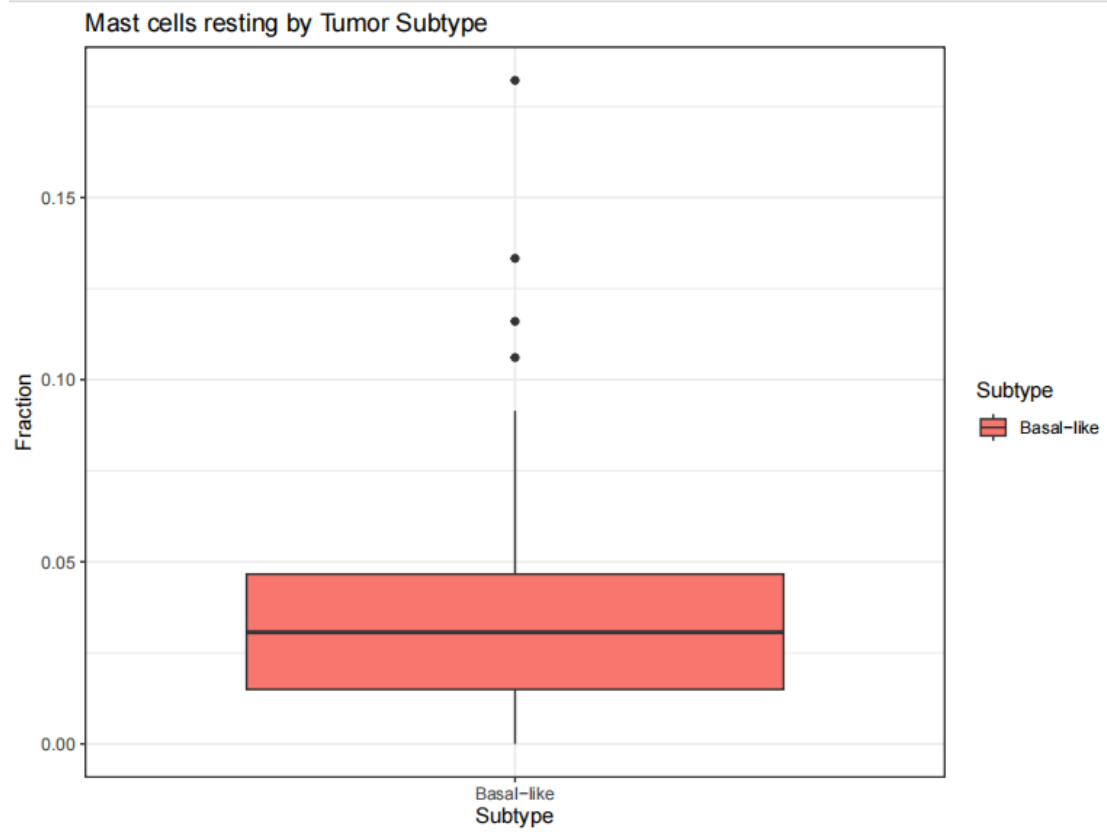


Fig 40: Mast cells resting by Tumor Subtype

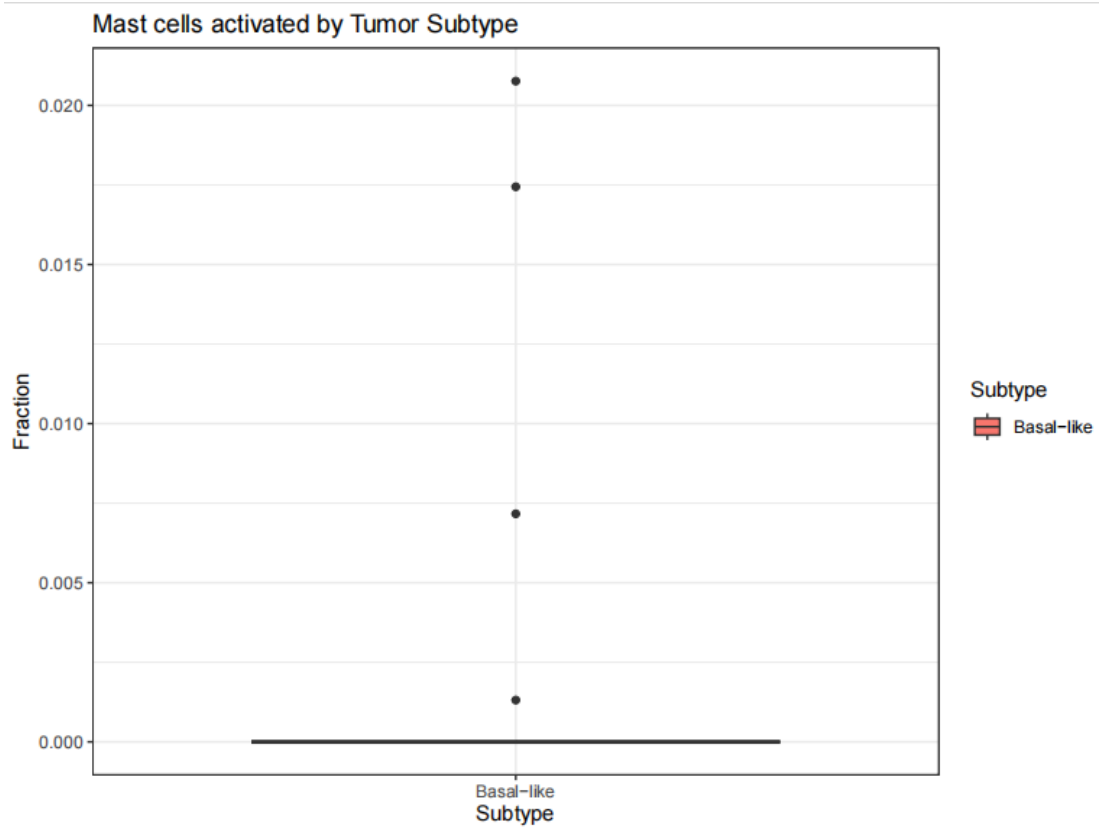


Fig 41: Mast cells activated by Tumor Subtype

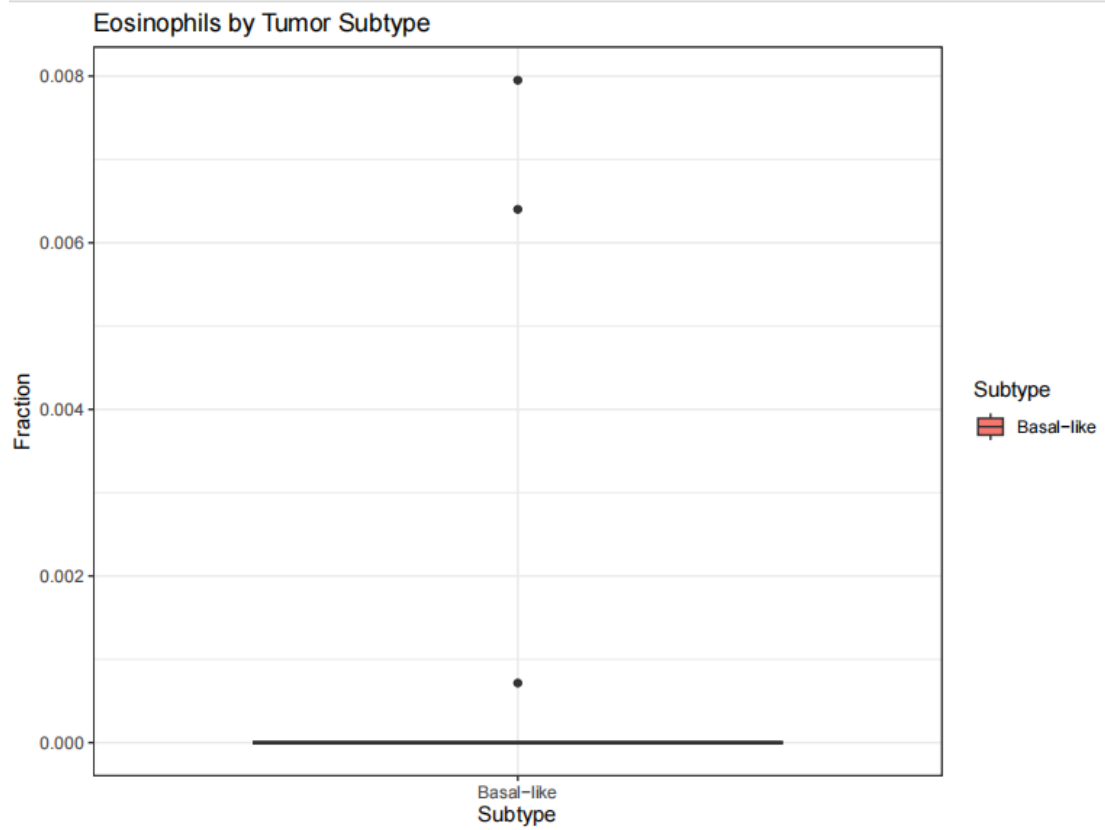


Fig 42: Eosinophils by Tumor Subtype

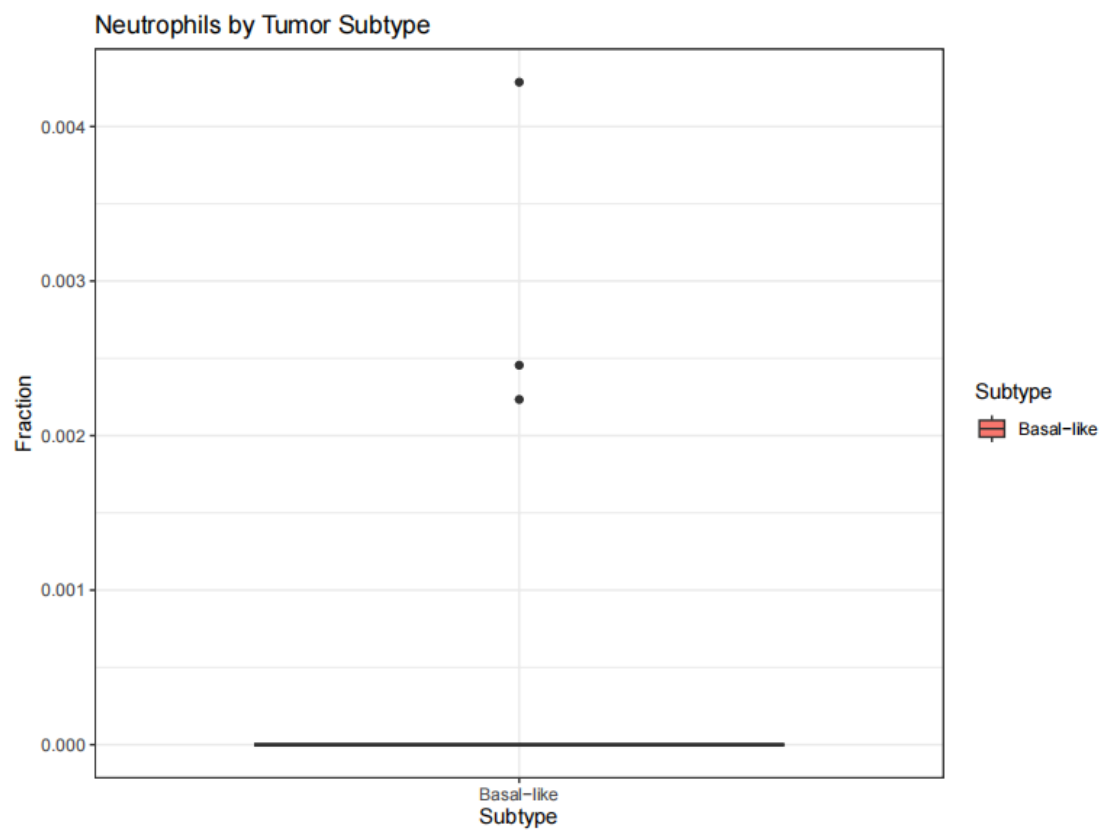


Fig 43: Neutrophils by Tumor Subtype

ANNEXURE C: R ANALYSIS PIPELINE:

This appendix contains the complete R analysis pipeline used for this study, organized by analytical stage. All scripts were executed in RStudio using R 4.4.2. Code comments reflect the iterative development process of the analysis.

C.1 Data Download and TNBC Sample Filtering

The TCGA-BRCA clinical matrix was downloaded directly from the UCSC Xena Browser. Samples were filtered for Triple-Negative Breast Cancer (ER–, PR–, HER2–) status using receptor annotations from the nature2012 dataset. The PAM50 molecular subtype was used to further identify the Basal-like subset within TNBC samples.

C.1.1 Loading the UCSCXenaTools Library and Updating Data

```
# install.packages("UCSCXenaTools")
library(UCSCXenaTools)

XenaDataUpdate()

# Generate TCGA Hub query
query <- XenaGenerate(subset = XenaHostNames == "tcgaHub")
```

C.1.2 Downloading the BRCA Clinical Matrix

```
# Make a data folder if it doesn't exist
if(!dir.exists("data")) dir.create("data")

# Download the clinical matrix directly from Xena
download.file(
  "https://tcga-xena-hub.s3.us-east-
1.amazonaws.com/download/TCGA.BRCA.sampleMap%2FBRCA_clinicalMatrix",
  destfile = "data/BRCA_clinicalMatrix.txt")
```

```
# Load and preview
pheno_data <- read.delim("data/BRCA_clinicalMatrix.txt",
                        header = TRUE, sep = "\t", stringsAsFactors = FALSE)

dim(pheno_data)
head(pheno_data[, 1:10])
colnames(pheno_data)
```

C.1.3 Filtering for TNBC and Basal-like Samples

```
library(dplyr)

# Filter TNBC: ER-, PR-, HER2-
tnbc_samples <- pheno_data %>%
  filter(ER_Status_nature2012 == "Negative",
         PR_Status_nature2012 == "Negative",
         HER2_Final_Status_nature2012 == "Negative")

# Subset that are also PAM50 Basal-like
tnbc_basal <- tnbc_samples %>%
  filter(PAM50_mRNA_nature2012 == "Basal-like")

# Counts
nrow(tnbc_samples) # TNBC by ER/PR/HER2
nrow(tnbc_basal)   # Basal-like subset

# PAM50 frequency table
pam50_df <- as.data.frame(table(tnbc_samples$PAM50_mRNA_nature2012))
pam50_df$Percent <- round(100 * pam50_df$Freq / sum(pam50_df$Freq), 2)
pam50_df

# Save sample info
tnbc_basal_selected <- tnbc_basal %>%
  select(sampleID, ER_Status_nature2012, PR_Status_nature2012,
```

```
HER2_Final_Status_nature2012, PAM50_mRNA_nature2012)
```

```
write.csv(tnbc_basal_selected,  
          "TNBC_full_sample_info.csv",  
          row.names = FALSE)
```

C.2 Expression Matrix Preparation

The TCGA-BRCA RNA-seq expression matrix ($\log_2(x+1)$ normalized RSEM counts, HiSeqV2) was downloaded and subset to the 76 TNBC Basal-like samples identified in C.1. Sample IDs were standardized to TCGA barcode format before subsetting.

C.2.1 Downloading the RNA-seq Expression Matrix

```
library(data.table)  
library(dplyr)  
library(readr)  
  
# Download HiSeqV2 expression data  
options(timeout = 600)  
expr_url <- "https://tcga-xena-hub.s3.us-east-  
1.amazonaws.com/download/TCGA.BRCA.sampleMap%2FHiSeqV2.gz"  
download.file(expr_url, destfile = "TCGA_BRCA_HiSeqV2.gz", mode = "wb")  
  
# Read expression file  
expr_data <- fread("TCGA_BRCA_HiSeqV2.gz")  
head(expr_data[, 1:5])  
nrow(expr_data) # ~20,531 genes  
ncol(expr_data) # 1218 samples + 1 gene column
```

C.2.2 Subsetting to TNBC Samples

```
# Load sample info
```

```
pheno <- read.csv("TNBC_full_sample_info.csv")

# Standardize IDs
pheno$sampleID <- gsub("\\.", "-", pheno$sampleID)

# Subset expression matrix to TNBC samples only
tnbc_expr <- expr_data[, c("sample", pheno$sampleID), with = FALSE]
dim(tnbc_expr)

# Save
write.csv(tnbc_expr, "TNBC_expression_matrix.csv", row.names = FALSE)
```

C.3 Principal Component Analysis and Heatmap

PCA was performed on the top 1000 most variable genes to assess transcriptomic structure within the TNBC cohort. Samples were colored by PD-L1 (CD274) expression group using a median-split. A heatmap of the same gene set was generated to visualize expression gradients.

C.3.1 Data Setup and Variance Filtering

```
library(data.table)
library(dplyr)
library(ggplot2)
library(pheatmap)

# Load data
expr_df <- fread("TNBC_expression_matrix.csv")
pheno_df <- read.csv("TNBC_full_sample_info.csv", stringsAsFactors = FALSE)
pheno_df$sampleID <- gsub("\\.", "-", pheno_df$sampleID)

# Rename gene column
colnames(expr_df)[1] <- "Gene"
```

```

# Match samples
common <- intersect(setdiff(colnames(expr_df), "Gene"), pheno_df$sampleID)
expr_df <- expr_df[, c("Gene", common), with = FALSE]
pheno_df <- pheno_df[match(common, pheno_df$sampleID), ]

# Build numeric matrix
expr_df <- expr_df[!is.na(expr_df$Gene) & expr_df$Gene != "", ]
expr_df <- expr_df[!duplicated(expr_df$Gene), ]
mat <- as.matrix(expr_df[, -1, with = FALSE])
mode(mat) <- "numeric"
rownames(mat) <- expr_df$Gene

# Top 1000 most variable genes
gene_var <- apply(mat, 1, var, na.rm = TRUE)
keep_genes <- names(sort(gene_var, decreasing = TRUE))[1:1000]
mat_var <- mat[keep_genes, ]

```

C.3.2 PD-L1 Median-Split Grouping

```

# Extract CD274 (PD-L1) expression
if ("CD274" %in% rownames(mat)) {
  cd274 <- mat["CD274", ]
  cutoff <- median(cd274, na.rm = TRUE)
  group <- ifelse(cd274 >= cutoff, "PD-L1_high", "PD-L1_low")
} else {
  message("CD274 not found in expression matrix.")
}

# Annotation data frame for heatmap
annot <- data.frame(
  PAM50 = pheno_df$PAM50_mRNA_nature2012,
  Group = group[pheno_df$sampleID],
  stringsAsFactors = FALSE
)

```

```
rownames(annot) <- pheno_df$sampleID
```

C.3.3 PCA Plot

```
# PCA (samples as rows)
pca <- prcomp(t(mat_var), center = TRUE, scale. = TRUE)
pca_df <- data.frame(
  Sample = rownames(pca$x),
  PC1    = pca$x[, 1],
  PC2    = pca$x[, 2],
  Group  = annot[rownames(pca$x), "Group"]
)

ggplot(pca_df, aes(PC1, PC2, color = Group)) +
  geom_point(size = 3, alpha = 0.9) +
  theme_minimal(base_size = 12) +
  labs(title = "PCA of Basal TNBC (top variable genes)",
       subtitle = "Colored by PD-L1 (CD274) median-split",
       color = "Group")

write.csv(pca_df, "TNBC_PCA_scores.csv", row.names = FALSE)
```

C.3.4 Heatmap

```
ann_colors <- list(
  Group = c("PD-L1_high" = "#1f77b4", "PD-L1_low" = "#ff7f0e"),
  PAM50 = c("Basal-like" = "#d62728")
)

pheatmap(mat_var,
  scale          = "row",
  show_rownames  = FALSE,
  show_colnames  = FALSE,
  clustering_distance_rows = "correlation",
```

```

clustering_distance_cols = "correlation",
clustering_method       = "ward.D2",
annotation_col          = annot,
annotation_colors       = ann_colors,
main                    = "Basal TNBC - Heatmap of Top Variable Genes")

```

C.4 Differential Gene Expression Analysis (limma-voom)

Differential expression between PD-L1-high and PD-L1-low TNBC groups was performed using the limma-voom framework. Genes with $|\log_2FC| > 1$ and FDR-adjusted p-value < 0.05 were considered significant. Results were visualized using volcano plots with top DEGs and immune checkpoint genes labelled.

C.4.1 PD-L1 Group Assignment and DEG Setup

```

library(limma)
library(edgeR)

# Assign PD-L1 groups based on CD274 median
pd_l1_values <- as.numeric(mat["CD274", ])
pheno_df$Group <- ifelse(pd_l1_values > median(pd_l1_values, na.rm = TRUE),
                        "PDL1_high", "PDL1_low")
table(pheno_df$Group)

```

C.4.2 limma-voom Pipeline

```

# DGEList object
dge <- DGEList(counts = mat)

# Design matrix
pheno_df$Group <- factor(pheno_df$Group, levels = c("PDL1_low", "PDL1_high"))
design <- model.matrix(~ Group, data = pheno_df)
colnames(design) <- c("Intercept", "PDL1_high_vs_low")

# voom transformation

```



```

v <- voom(dge, design, plot = TRUE)

# Fit linear model and apply empirical Bayes
fit <- lmFit(v, design)
fit <- eBayes(fit)

# Extract results
deg_results <- topTable(fit,
                        coef      = "PDL1_high_vs_low",
                        number     = Inf,
                        adjust.method = "BH")

write.csv(deg_results, "TNBC_PDL1_DEG_results.csv", row.names = TRUE)
summary(decideTests(fit))

```

C.4.3 Volcano Plot with Gene Labels

```

library(ggplot2)
library(ggrepel)

logFC_cutoff <- 1
adjP_cutoff <- 0.05

deg <- topTable(fit, coef = "PDL1_high_vs_low", number = Inf, sort.by = "P")

# Significance classification
deg$Significance <- "Not Sig"
deg$Significance[deg$logFC >= logFC_cutoff & deg$adj.P.Val < adjP_cutoff] <- "Up"
deg$Significance[deg$logFC <= -logFC_cutoff & deg$adj.P.Val < adjP_cutoff] <-
"Down"

# Genes to label: top 10 significant + immune checkpoint genes
top_sig <- deg[deg$Significance != "Not Sig", ]
top_sig <- top_sig[order(top_sig$adj.P.Val), ][1:10, ]

```

```

checkpoints          <-          deg[rownames(deg)          %in%
c("CD274","PDCD1","CTLA4","LAG3","TIGIT"), ]
label_genes  <- rbind(top_sig, checkpoints)

# Plot
volcano <- ggplot(deg, aes(x = logFC, y = -log10(adj.P.Val), color = Significance)) +
  geom_point(alpha = 0.6, size = 1.5) +
  scale_color_manual(values = c("Down" = "blue", "Not Sig" = "grey", "Up" = "red")) +
  geom_vline(xintercept = c(-logFC_cutoff, logFC_cutoff), linetype = "dashed") +
  geom_hline(yintercept = -log10(adjP_cutoff), linetype = "dashed") +
  geom_text_repel(data = label_genes,
                  aes(label = rownames(label_genes)),
                  size = 3, max.overlaps = 50) +
  labs(title = "Volcano Plot of DEGs (PD-L1 High vs Low)",
       x = "Log2 Fold Change",
       y = "-Log10 Adjusted P-value") +
  theme_minimal()

ggsave("Volcano_Plot_DEGs.png", volcano, width = 8, height = 6, dpi = 300)

# Export DEG tables
write.csv(subset(deg, Significance == "Up"), "Upregulated_genes.csv")
write.csv(subset(deg, Significance == "Down"), "Downregulated_genes.csv")
write.csv(deg, "All_DEGs.csv")

```

C.5 Tumor Mutational Burden (TMB) Calculation

TMB was calculated from the TCGA-BRCA MC3 somatic mutation MAF file. Only coding, high-confidence variant classes were retained. Mutation counts were normalized per 38 Mb of the sequenced exome to yield mutations per megabase (mut/Mb).

C.5.1 Downloading and Processing the MAF File

```
library(data.table)
library(dplyr)
library(ggplot2)

# Download BRCA MC3 MAF
maf_url <- "https://tcga-xena-hub.s3.us-east-1.amazonaws.com/download/mc3%2FBRCA_mc3.txt.gz"
maf <- fread(cmd = paste("curl -s", maf_url))
colnames(maf)
```

C.5.2 Computing TMB per Sample

```
# Count all mutations per sample and normalize
tmb_counts <- maf %>%
  group_by(sample) %>%
  summarise(Mutation_Count = n()) %>%
  mutate(TMB = Mutation_Count / 38) # 38 Mb exome

write.csv(tmb_counts, "BRCA_TMB.csv", row.names = FALSE)
```

C.5.3 Visualizing TMB Distribution

```
ggplot(tmb_counts, aes(x = TMB)) +
  geom_histogram(binwidth = 0.2, fill = "steelblue", color = "black") +
  theme_minimal() +
  labs(title = "TMB Distribution in TCGA-BRCA Cohort",
       x = "Tumor Mutational Burden (mutations/Mb)",
       y = "Number of Samples")
```

C.6 CIBERSORTx Input Preparation

The TNBC expression matrix was reformatted as a tab-delimited text file compatible with CIBERSORTx. Duplicate gene symbols were resolved by retaining the row with the highest median expression. Genes with all-zero values were removed prior to submission.

C.6.1 Formatting the Expression File

```
library(dplyr)
library(readr)

# Read expression matrix
expr_raw <- read_csv("TNBC_expression_matrix.csv")
colnames(expr_raw)[1] <- "GeneSymbol"

# Remove empty gene symbols
expr_clean <- expr_raw %>%
  filter(!is.na(GeneSymbol), GeneSymbol != "")

# Keep row with highest median per duplicate gene
expr_final <- expr_clean %>%
  rowwise() %>%
  mutate(.med = median(c_across(-GeneSymbol), na.rm = TRUE)) %>%
  ungroup() %>%
  arrange(GeneSymbol, desc(.med)) %>%
  distinct(GeneSymbol, .keep_all = TRUE) %>%
  select(-.med)

# Save as tab-delimited for CIBERSORTx upload
write.table(expr_final,
  file = "TNBC_expression_CIBERSORTx_ready.txt",
  sep = "\t",
  row.names = FALSE,
  quote = FALSE)
```

```
# Verify output
test <- read.delim("TNBC_expression_CIBERSORTx_ready.txt", check.names = FALSE)
dim(test)
head(test[, 1:5])
```

C.7 Integrated Analysis: CIBERSORTx, PD-L1, TMB, and Clinical Data

CIBERSORTx immune cell fraction estimates were merged with PD-L1 expression, TMB values, and TCGA clinical metadata. A composite Tumor-Reactive Score (TRS) was derived from z-scored PD-L1, CD8+ T cell, and log-TMB values. Spearman correlations, Kaplan-Meier survival analysis, and IDO2-specific immune correlations were performed on the integrated dataset.

C.7.1 Loading and Harmonizing Sample IDs

```
library(dplyr)
library(data.table)

# Load files
immune <- read.csv("CIBERSORTx_Job1_Results.csv", row.names = 1)
expr <- read.csv("TNBC_expression_matrix.csv", row.names = 1)
tmb <- read.csv("BRCA_TMB.csv")
clin <- read.table("BRCA_clinicalMatrix.txt",
                  header = TRUE, sep = "\t",
                  quote = "", stringsAsFactors = FALSE)

# Extract PD-L1 expression per sample
cd274_expr <- as.data.frame(t(expr["CD274", ]))
colnames(cd274_expr) <- "CD274"

# Standardize all IDs to first 12 characters of TCGA barcode
clean_id <- function(x) substr(gsub("\\.", "-", x), 1, 12)
```

```

rownames(cd274_expr) <- clean_id(rownames(cd274_expr))
rownames(immune) <- clean_id(rownames(immune))
tmb$SampleID <- clean_id(tmb$sample)
clin$SampleID <- clean_id(clin$sampleID)

cd274_expr$SampleID <- rownames(cd274_expr)
immune$SampleID <- rownames(immune)

```

C.7.2 Merging Datasets

```

# Sequential merge
merged_data <- merge(cd274_expr, immune, by = "SampleID")
merged_data <- merge(merged_data, tmb, by = "SampleID")
merged_data <- merge(merged_data, clin, by = "SampleID")
dim(merged_data)
colSums(is.na(merged_data))

# Select key variables for analysis
analysis_df <- merged_data[, c(
  "SampleID", "CD274", "TMB",
  "T.cells.CD8",
  "OS_Time_nature2012", "OS_event_nature2012",
  "age_at_initial_pathologic_diagnosis",
  "pathologic_stage"
)]

```

C.7.3 Spearman Correlation Analysis

```

# PD-L1 vs CD8+ T cells
cor.test(analysis_df$CD274, analysis_df$T.cells.CD8, method = "spearman")

# PD-L1 vs TMB
cor.test(analysis_df$CD274, analysis_df$TMB, method = "spearman")

```

C.7.4 Composite Tumor-Reactive Score (TRS)

```
# Log-transform TMB to reduce skew
analysis_df$TMB_log <- log2(analysis_df$TMB + 1)

# Z-score each component
analysis_df$PDL1_z <- scale(analysis_df$CD274)
analysis_df$CD8_z <- scale(analysis_df$T.cells.CD8)
analysis_df$TMB_z <- scale(analysis_df$TMB_log)

# Composite TRS
analysis_df$TRS <- analysis_df$PDL1_z +
  analysis_df$CD8_z +
  analysis_df$TMB_z

summary(analysis_df$TRS)
hist(analysis_df$TRS)

analysis_df$TRS_group <- ifelse(
  analysis_df$TRS > median(analysis_df$TRS),
  "Immune-High", "Immune-Low")
table(analysis_df$TRS_group)
```

C.7.5 Kaplan-Meier Survival Analysis

```
library(survival)
library(survminer)

# PD-L1 high vs low survival
analysis_df$PDL1_group <- ifelse(
  analysis_df$CD274 > median(analysis_df$CD274),
  "High", "Low")
```

```

fit_pdl1 <- survfit(
  Surv(OS_Time_nature2012, OS_event_nature2012) ~ PDL1_group,
  data = analysis_df)

km_plot <- ggsurvplot(fit_pdl1, data = analysis_df, pval = TRUE)
print(km_plot)
ggsave("km_plot.png", width = 6, height = 4, dpi = 300)

# TRS tertile survival
analysis_df$TRS_group3 <- cut(
  analysis_df$TRS,
  breaks = quantile(analysis_df$TRS, probs = c(0, 0.33, 0.66, 1)),
  labels = c("Low", "Medium", "High"),
  include.lowest = TRUE)

fit3 <- survfit(
  Surv(OS_Time_nature2012, OS_event_nature2012) ~ TRS_group3,
  data = analysis_df)

ggsurvplot(fit3, data = analysis_df, pval = TRUE)
ggsave("fit3.png", width = 6, height = 4, dpi = 300)

```

C.7.6 Spearman Correlation Heatmap

```

library(pheatmap)

vars <- analysis_df[, c("CD274", "T.cells.CD8", "TMB_log", "TRS")]
cor_matrix <- cor(vars, method = "spearman")

pheatmap(cor_matrix, display_numbers = TRUE)

```

C.7.7 IDO2 Expression and Immune Correlations

```

# Extract IDO2 expression

```



```

ido2_df <- data.frame(
  SampleID = colnames(expr),
  IDO2     = as.numeric(expr["IDO2", ])
)
ido2_df$SampleID <- clean_id(gsub("\\.", "-", ido2_df$SampleID))

# Merge with analysis_df
merged_df <- merge(analysis_df, ido2_df, by = "SampleID")

# Add Treg and M2 fractions
immune_subset <- data.frame(
  SampleID = clean_id(rownames(cibersort)),
  Tregs    = cibersort$T.cells.regulatory..Tregs.,
  M2       = cibersort$Macrophages.M2
)
merged_df <- merge(merged_df, immune_subset, by = "SampleID")

# Correlations
cor.test(merged_df$IDO2, merged_df$Tregs, method = "spearman")
cor.test(merged_df$IDO2, merged_df$M2, method = "spearman")

# IDO2 vs Tregs scatterplot
ggplot(merged_df, aes(x = IDO2, y = Tregs)) +
  geom_point() +
  geom_smooth(method = "lm") +
  labs(title = "IDO2 vs Tregs")
ggsave("IDO2_vs_Tregs.png", width = 6, height = 4, dpi = 300)

# IDO2 KM survival
merged_df$IDO2_group <- ifelse(
  merged_df$IDO2 > median(merged_df$IDO2, na.rm = TRUE),
  "High", "Low")

fit_ido2 <- survfit(

```

```
Surv(OS_Time_nature2012, OS_event_nature2012) ~ IDO2_group,  
data = merged_df)  
  
p_ido2 <- ggsurvplot(fit_ido2, data = merged_df,  
                    pval = TRUE, risk.table = TRUE, conf.int = TRUE)  
png("KM_IDO2.png", width = 800, height = 600)  
print(p_ido2)  
dev.off()
```

ANNEXURE D:

D.1. Voom:Mean-variance trend

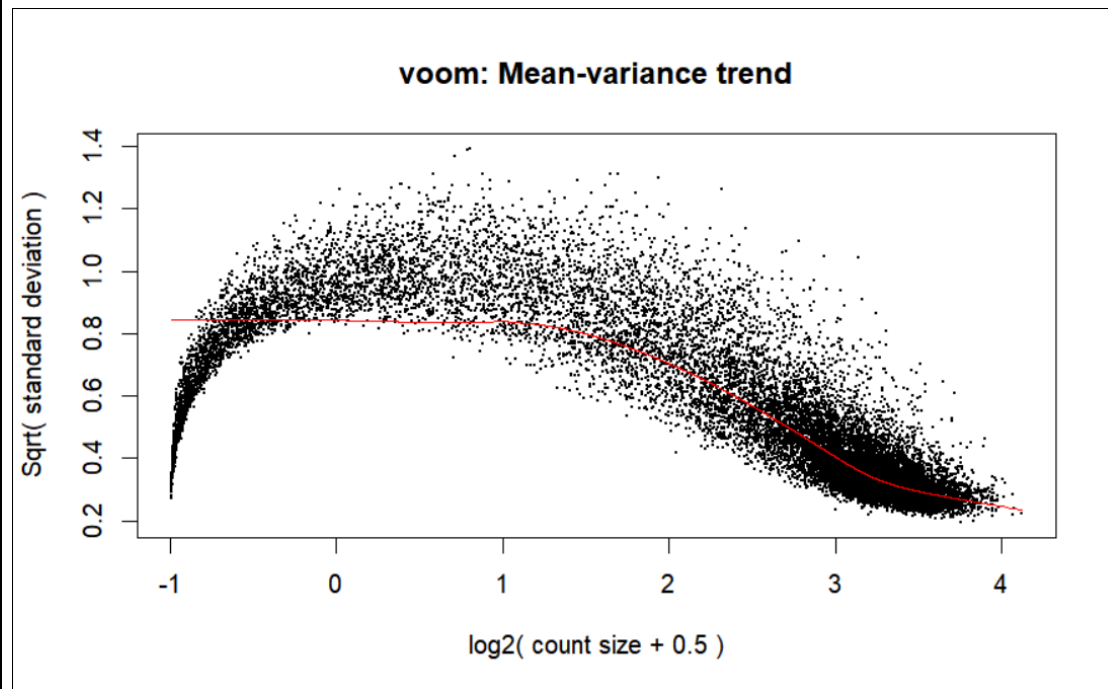


Fig 44: Voom Mean-Variance Trend Plot for TNBC Expression Data Prior to Differential Expression Analysis

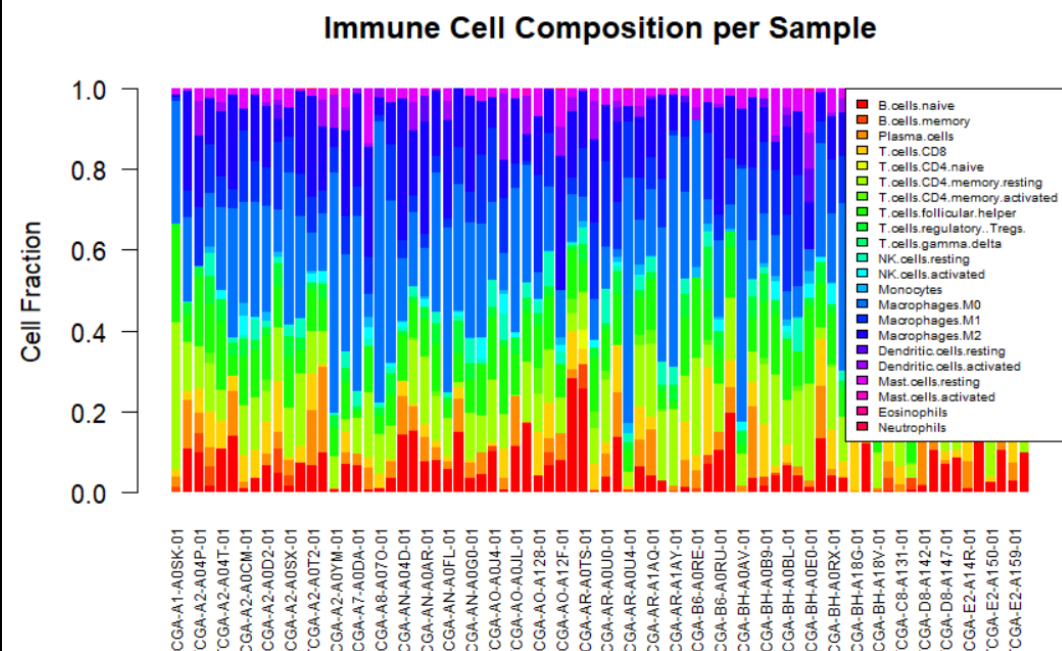


Fig 45: Stacked barplot of Immune Cell composition per Sample

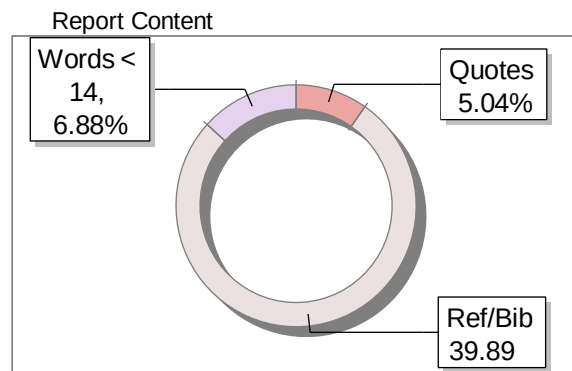
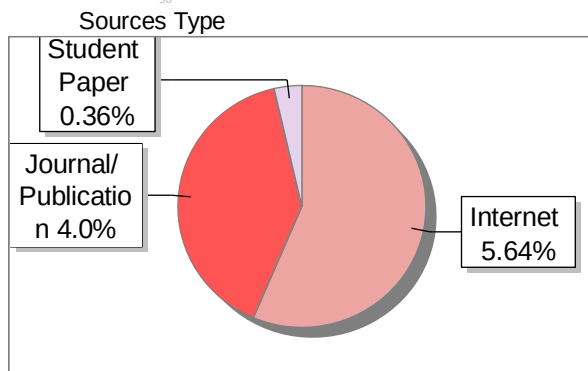
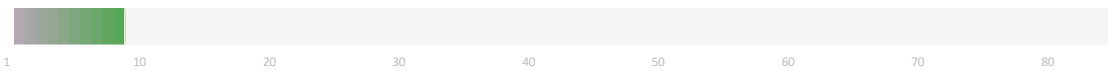
The Report is Generated by DrillBit Plagiarism Detection Software

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